

Lot-Sizing Recommendations - FY2027

Delta Pumps & Valves Co. (fictional case)

Generated 28 September 2026 | Prepared by Industrial Engineering Intern | Lot-Sizing Tool v1.0.0

Items analysed

22

Annual savings vs current

\$41,700

Avg inventory value (recommended)

\$286,172

Avg inventory value (current)

\$309,125

Orders per year (recommended)

130

Orders per year (current)

180

Savings, current cost and current inventory refer only to items for which a current order quantity was supplied.

Models: EOQ (Wilson), EPQ (finite production rate), all-units quantity discounts. Deterministic, level demand.

How to read this report

Delta Pumps & Valves Co. (fictional case) | 22 items | currency \$

What was calculated

For every item the tool finds the order (or production) quantity Q that minimises the total annual cost = ordering/setup cost ($A \cdot D/Q$) + carrying cost ($Q/2 \cdot v \cdot r$, multiplied by $(1-D/m)$ for produced items) + purchase cost ($D \cdot v$, which only matters when quantity discounts exist).

Models used

EOQ = $\sqrt{2AD / vr}$ for purchased items (entire order arrives at once).

EPQ = $\sqrt{2AD / (vr(1-D/m))}$ for items produced in-house at rate m .

All-units quantity discounts: the optimum of each price tier is found; if it falls below the tier's break quantity the break quantity is used; all tiers are then compared on total cost (lecture Cases 1, 2 and 3).

Practical rules

The theoretical optimum is rounded to the pack size and kept within the minimum and maximum order quantities. The column 'Practical rules applied' shows which rules shaped each answer.

Reorder point

Place an order when the inventory position (on hand + on order) falls to $D \times L$, where L is the lead time. Demand is assumed level and known; safety stock for uncertain demand is NOT included.

Flexibility band

Any order size between 'Flex band low' and 'Flex band high' costs at most 2% more ordering+carrying cost than the optimum. The EOQ cost curve is flat near its minimum ($PCP = 50 a^2/(1+a)$), so rounding to pallets or truckloads inside this band is essentially free.

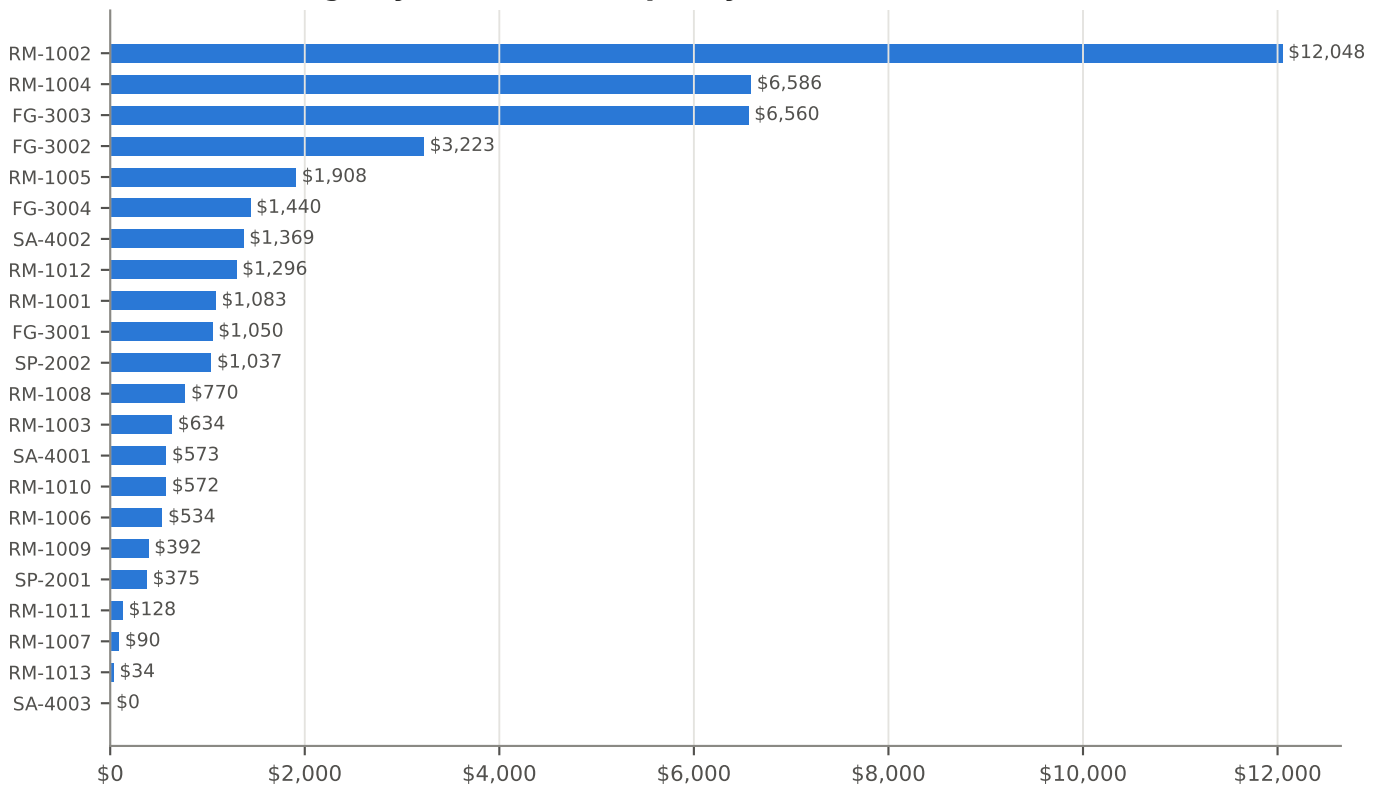
PCP of the current policy

Percentage Cost Penalty = $(\text{cost of current } Q - \text{cost of recommended } Q) / \text{optimal TRC} \times 100$.

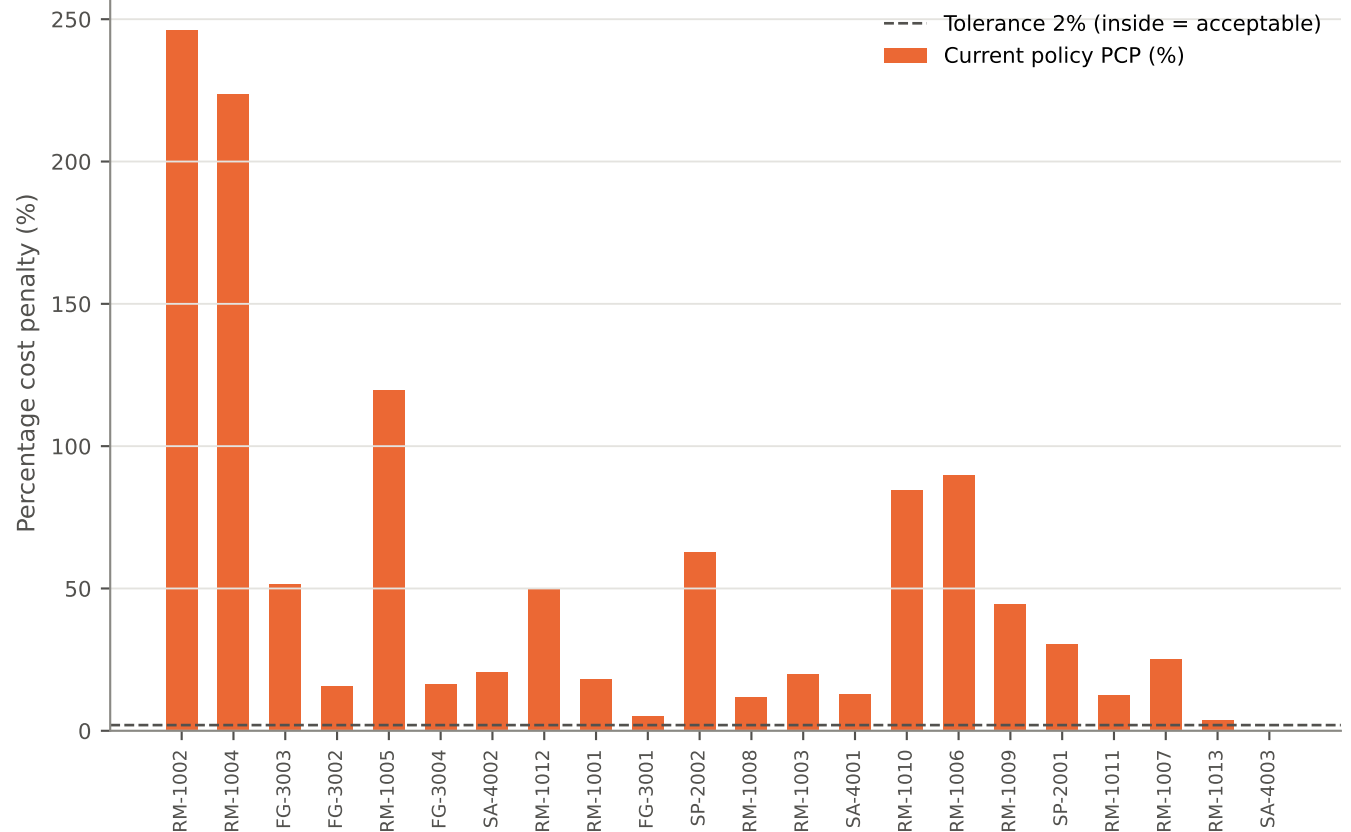
Limitations

Level, deterministic demand; costs constant over time; items treated independently (no joint replenishment); no shortages. Re-run the analysis when inputs change significantly.

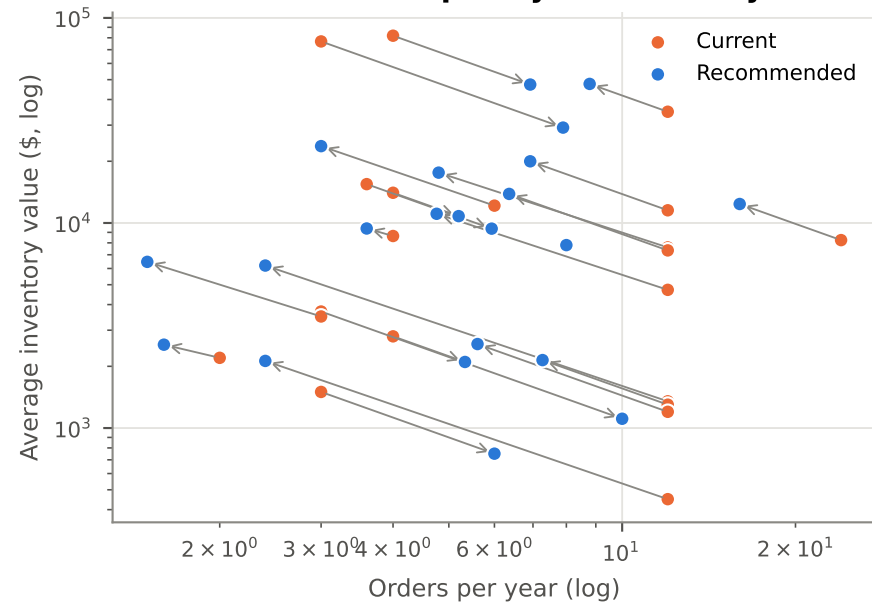
Annual savings by SKU (current policy -> recommended), total \$41,700



How far is each current policy from optimal? (PCP, SKUs ordered by savings)



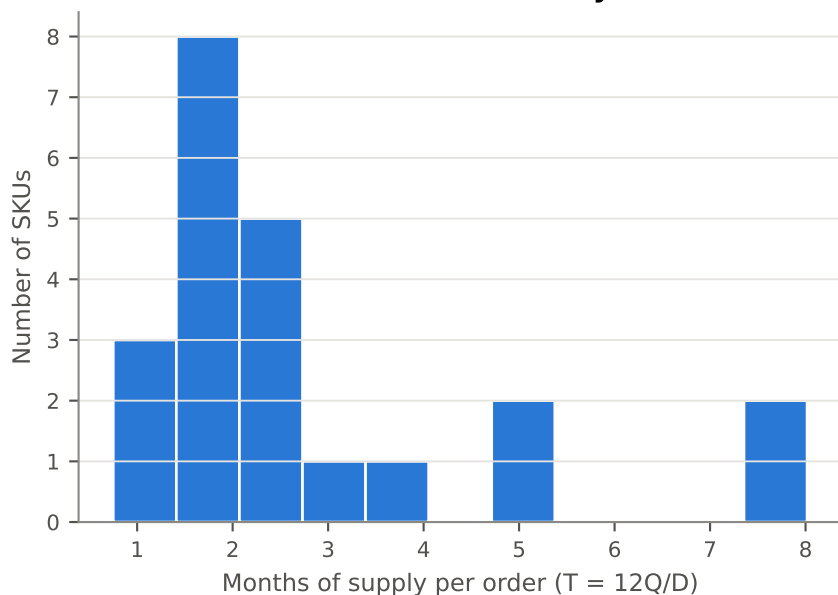
Trade-off: order frequency vs inventory



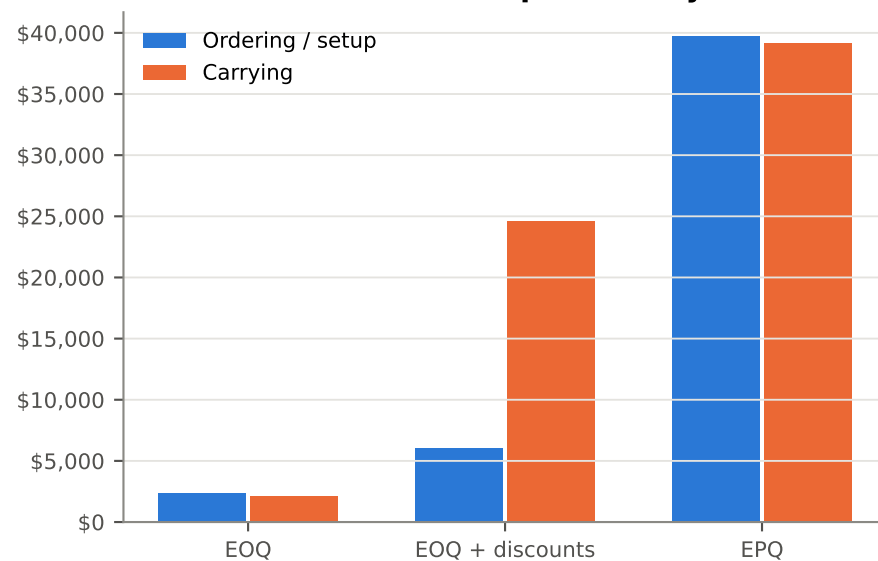
Average inventory investment (items with current policy)



Distribution of recommended cycle times



Recommended TRC components by model



Recommendations - summary table

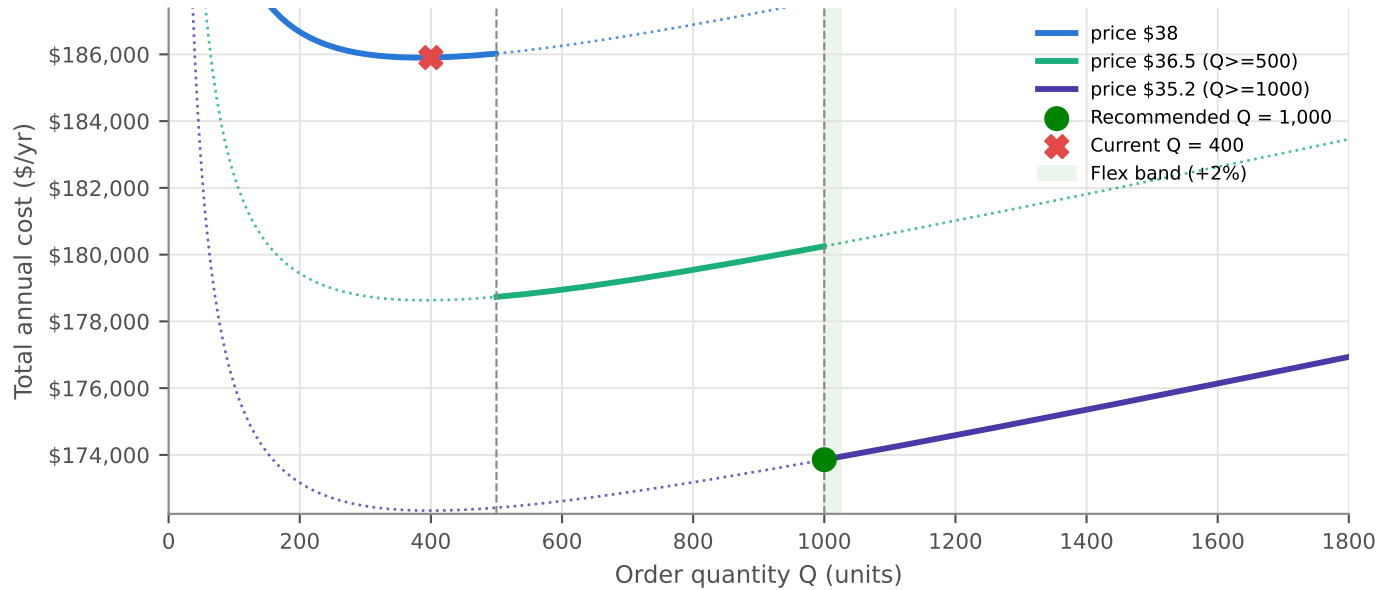
SKU	Model	Rec. Q	Textbook Q	Price (\$)	Orders/yr	Cycle (days)	ROP	Avg inv (\$)	TRC (\$)	Current Q	Savings (\$)	PCP %
RM-1001	EOQ + discount	12,000	2,464.0	3.950	3.0	83.3	1,440	23,700	5,943	6,000	1,083	18.2
RM-1002	EOQ + discount	1,000	383.9	35.200	4.8	52.1	384	17,600	4,896	400	12,048	246.1
RM-1003	EOQ + discount	1,000	480.7	21.600	5.2	48.1	312	10,800	3,216	1,300	634	19.7
RM-1004	EOQ + discount	2,016	637.8	11.000	4.8	52.5	461	11,088	2,947	800	6,586	223.5
RM-1005	EOQ + discount	10,000	2,582.0	1.240	2.4	104.2	672	6,200	1,596	2,000	1,908	119.5
RM-1006	EOQ + discount	25,000	9,860.1	0.170	2.4	104.2	1,200	2,125	594	5,000	534	89.9
RM-1007	EOQ	25,000	25,000.0	0.060	6.0	41.7	3,000	750	360	50,000	90	25.0
RM-1008	EOQ + discount	150	163.3	165.000	16.0	15.6	288	12,375	6,490	100	770	11.9
RM-1009	EOQ	300	385.5	7.400	10.0	25.0	96	1,110	883	1,000	392	44.4
RM-1010	EOQ + discount	5,000	1,557.0	1.020	1.6	156.2	192	2,550	676	4,000	572	84.6
RM-1011	EOQ	165	164.1	26.000	7.3	34.4	67	2,145	1,024	100	128	12.5
RM-1012	EOQ + discount	2,000	770.6	9.400	3.6	69.4	518	9,400	2,598	1,800	1,296	49.9
SP-2001	EOQ	107	107.0	48.000	5.6	44.6	60	2,568	1,233	50	375	30.4
SP-2002	EOQ + discount	1,002	240.9	12.900	1.5	167.0	90	6,463	1,648	500	1,037	62.9
FG-3001	EPQ	410	411.0	310.000	8.8	28.5	72	47,662	21,022	300	1,050	5.0
FG-3002	EPQ	260	259.7	455.000	6.9	36.1	36	47,320	20,795	450	3,223	15.5
FG-3003	EPQ	760	753.8	96.000	7.9	31.7	96	29,184	12,736	2,000	6,560	51.5
FG-3004	EPQ	1,300	1,315.1	41.000	6.9	36.1	144	19,988	8,897	750	1,440	16.2
SA-4001	EPQ	910	910.5	27.500	5.9	42.1	65	9,384	4,507	1,500	573	12.7
SA-4002	EPQ	660	658.3	63.000	6.4	39.3	50	13,860	6,635	350	1,369	20.6
SA-4003	EPQ	600	679.4	52.000	8.0	31.2	38	7,800	4,272	600	0	0.0
RM-1013	EOQ	375	382.2	11.200	5.3	46.9	160	2,100	942	500	34	3.6

Input validation log

sku	level	message
RM-1013	WARNING	carrying_rate 22.0 > 1 - did you enter a percentage? Interpreted as 0.22
FG-3005	ERROR	production_rate (2000) must exceed annual_demand (2500) - item skipped

RM-1002 - Cast iron pump housing [EOQ + discounts]

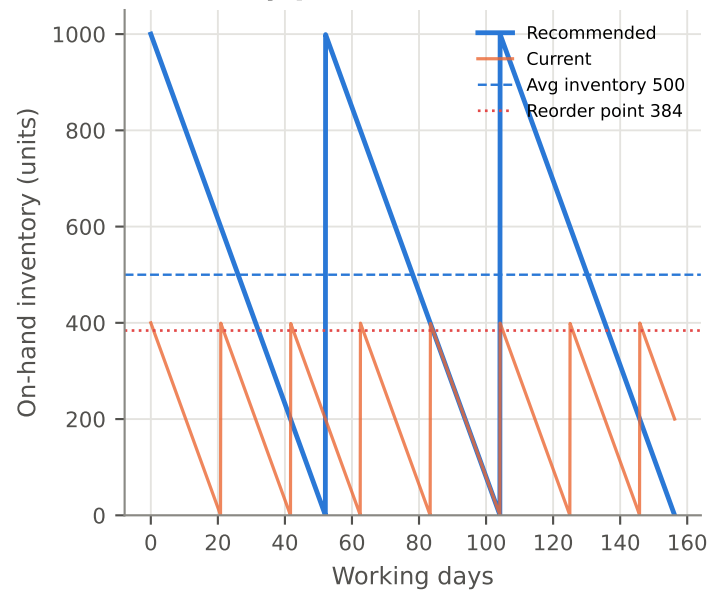
Total annual cost curve (solid = applicable price, dotted = extension)



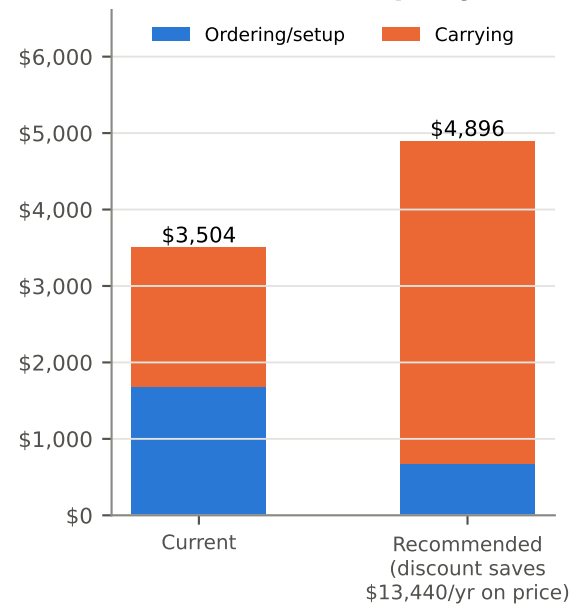
Key figures

Metric	Value
Annual demand D	4,800
Order/setup cost A	\$140.00
Base unit cost v0	\$38.000
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	383.9
Recommended Q	1,000
Price paid	\$35.200
Orders per year	4.8
Cycle (working days)	52.1
Reorder point	384
TRC (order+carry)	\$4,896
Total annual cost	\$173,856
Case / tier	Tier 3 of 3
Rules applied	Pack 10
Current Q	400
Annual savings	\$12,048
Current PCP	246.1%

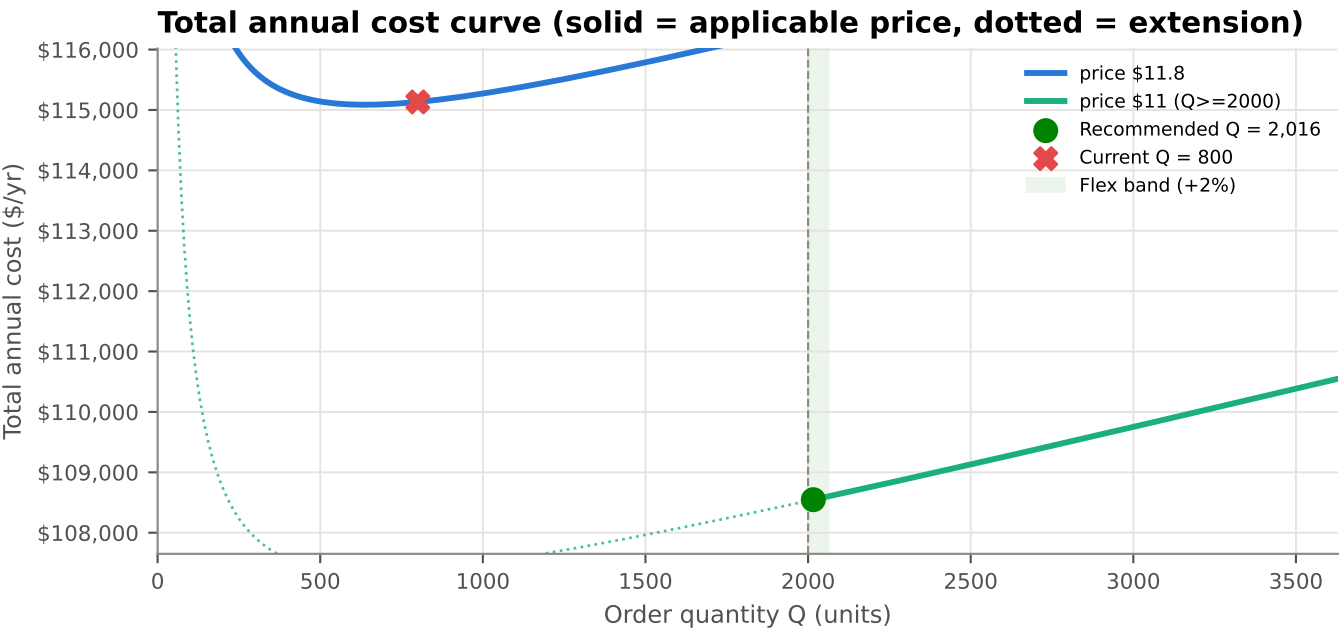
Inventory profile over time (sawtooth)



TRC breakdown (per year)

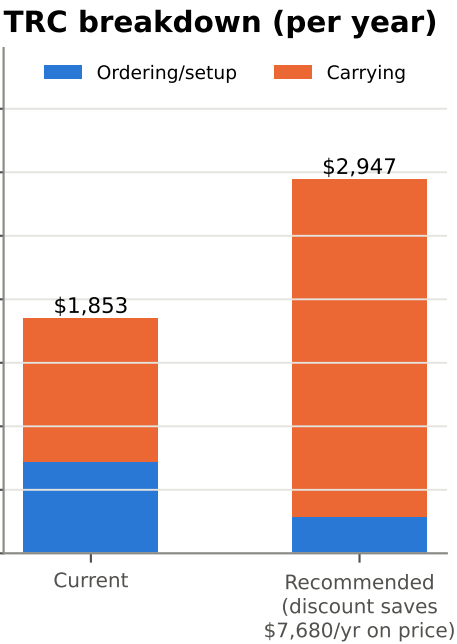
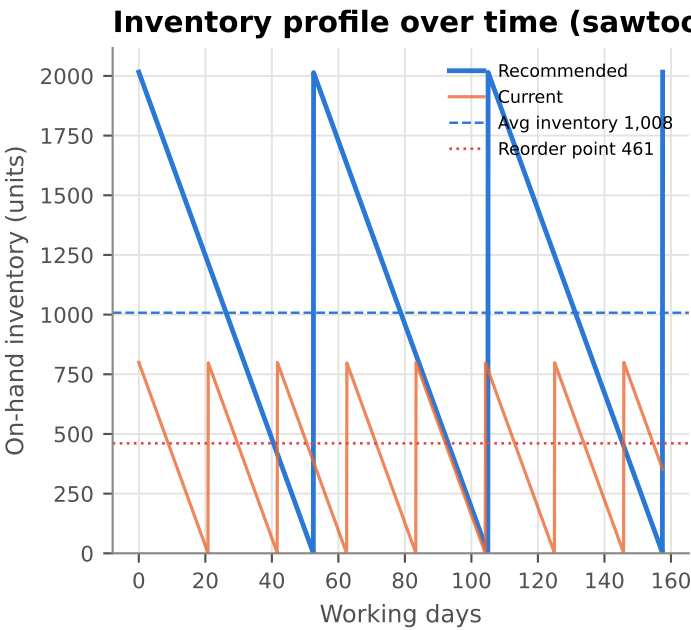


RM-1004 - Mechanical seal kit [EOQ + discounts]



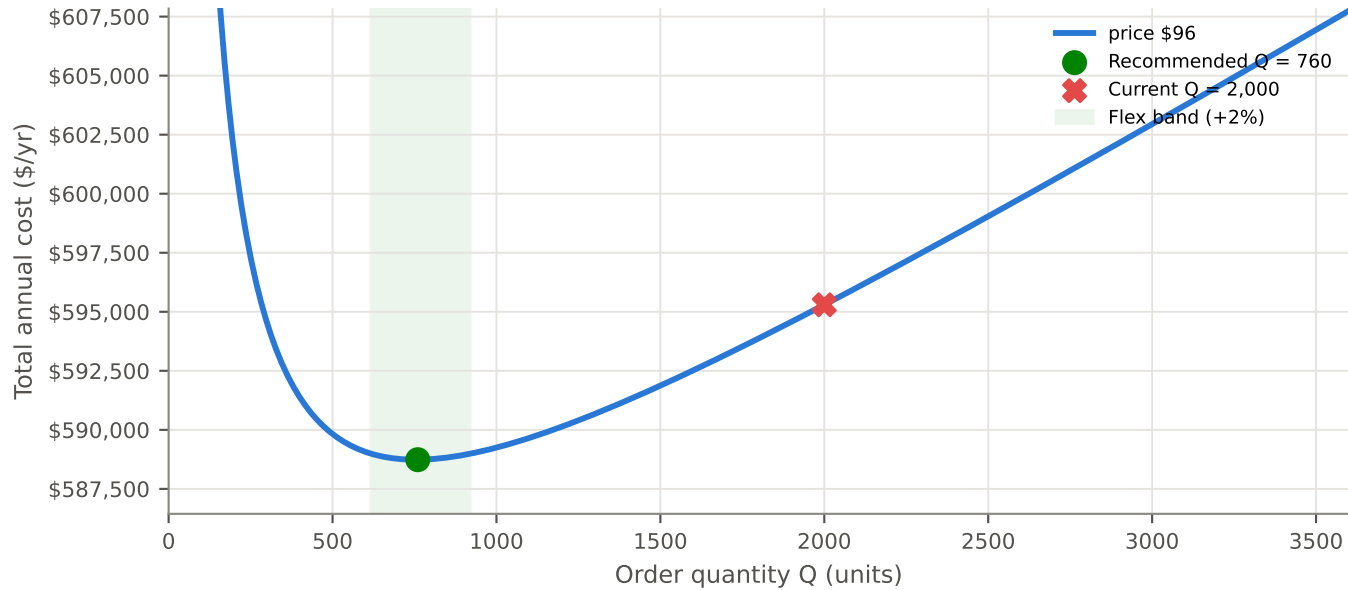
Key figures

Metric	Value
Annual demand D	9,600
Order/setup cost A	\$60.00
Base unit cost v0	\$11.800
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	637.8
Recommended Q	2,016
Price paid	\$11.000
Orders per year	4.8
Cycle (working days)	52.5
Reorder point	461
TRC (order+carry)	\$2,947
Total annual cost	\$108,547
Case / tier	Case 1
Rules applied	Pack 24
Current Q	800
Annual savings	\$6,586
Current PCP	223.5%

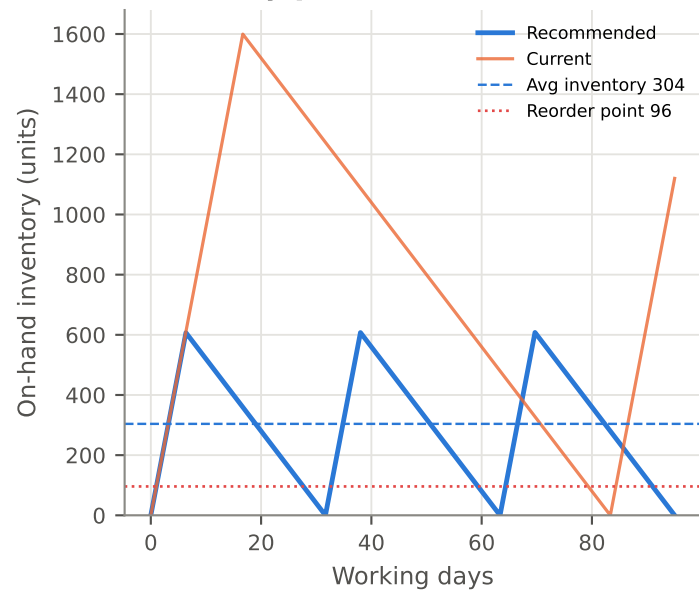


FG-3003 - Gate valve DN100 [EPQ]

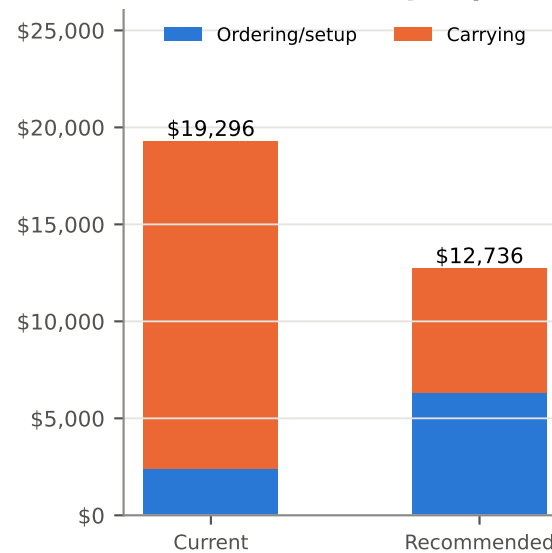
Total annual cost curve (solid = applicable price, dotted = extension)



Inventory profile over time (sawtooth)



TRC breakdown (per year)

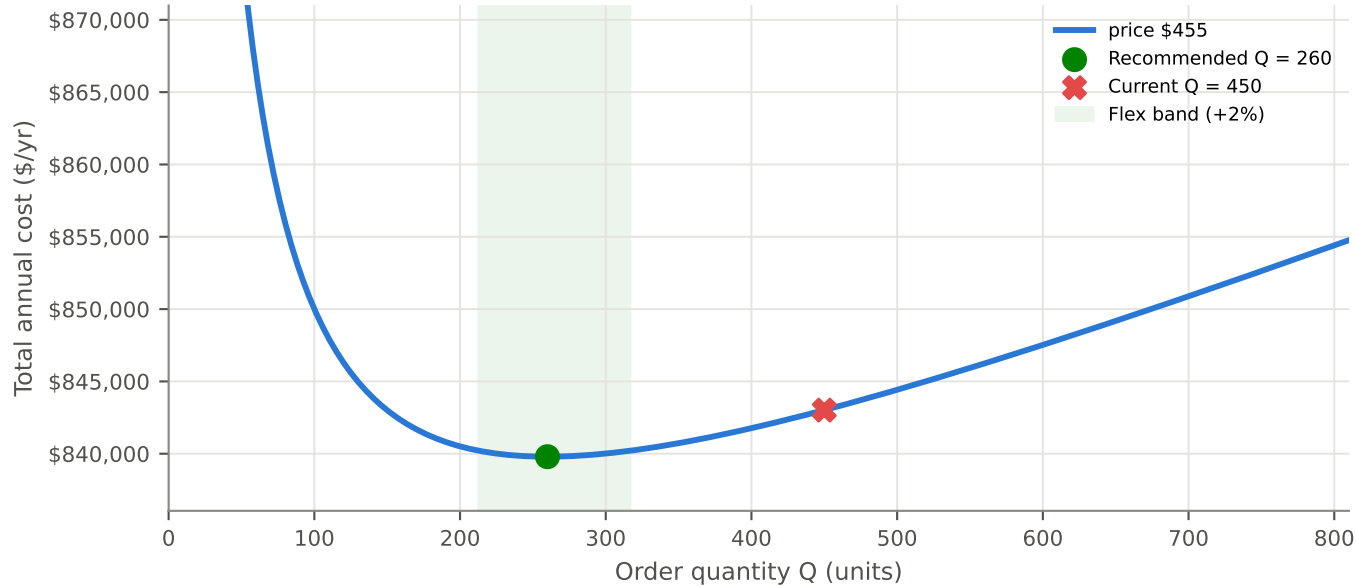


Key figures

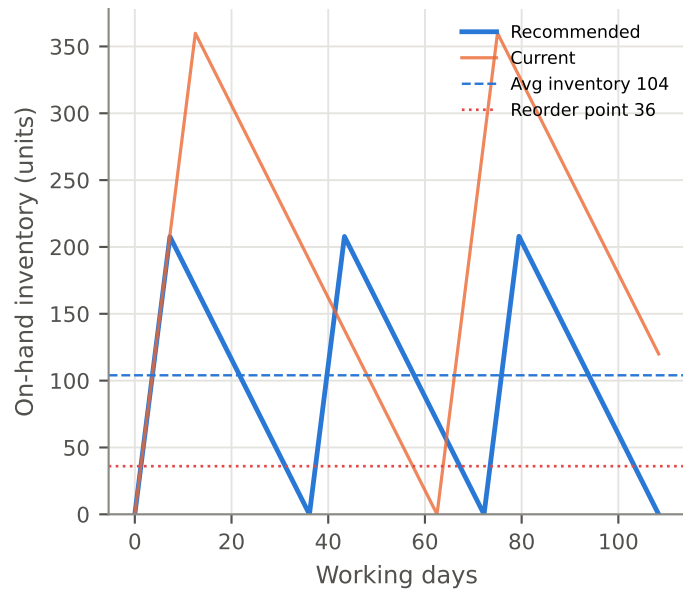
Metric	Value
Annual demand D	6,000
Order/setup cost A	\$800.00
Base unit cost v0	\$96.000
Carrying rate r	0.22
Production rate m	30,000
Textbook EOQ/EPQ	753.8
Recommended Q	760
Price paid	\$96.000
Orders per year	7.9
Cycle (working days)	31.7
Reorder point	96
TRC (order+carry)	\$12,736
Total annual cost	\$588,736
Case / tier	-
Rules applied	Pack 20
Current Q	2,000
Annual savings	\$6,560
Current PCP	51.5%

FG-3002 - Centrifugal pump CP-80 [EPQ]

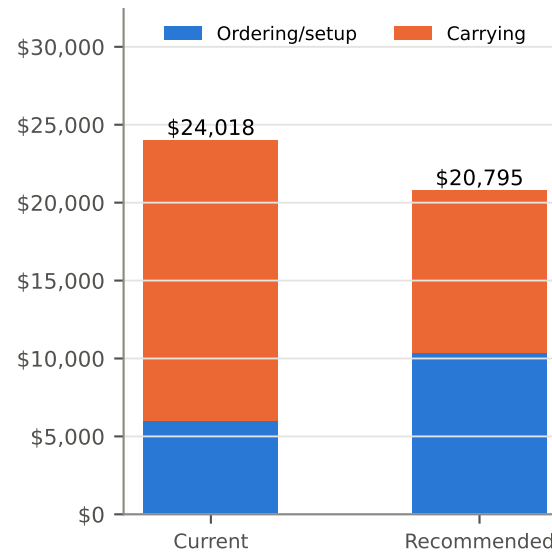
Total annual cost curve (solid = applicable price, dotted = extension)



Inventory profile over time (sawtooth)



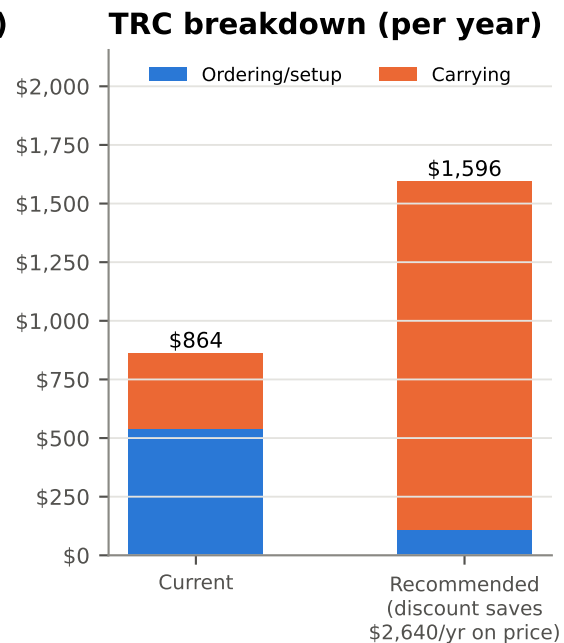
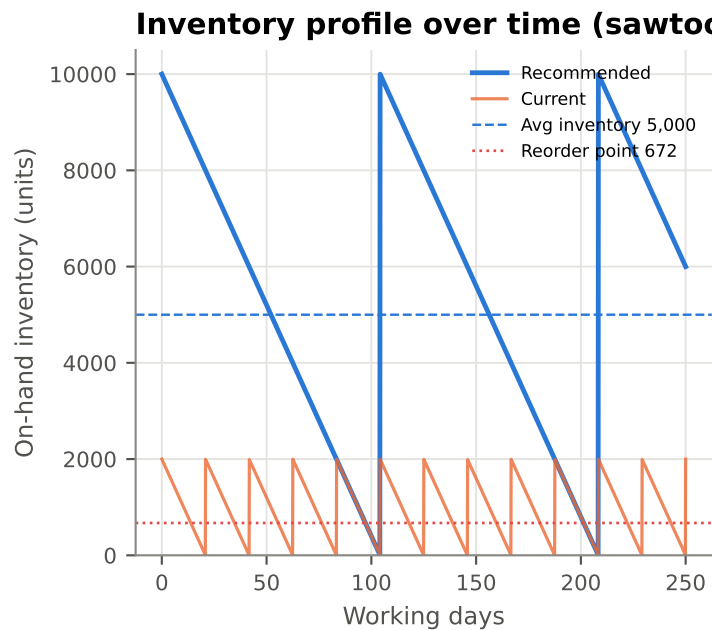
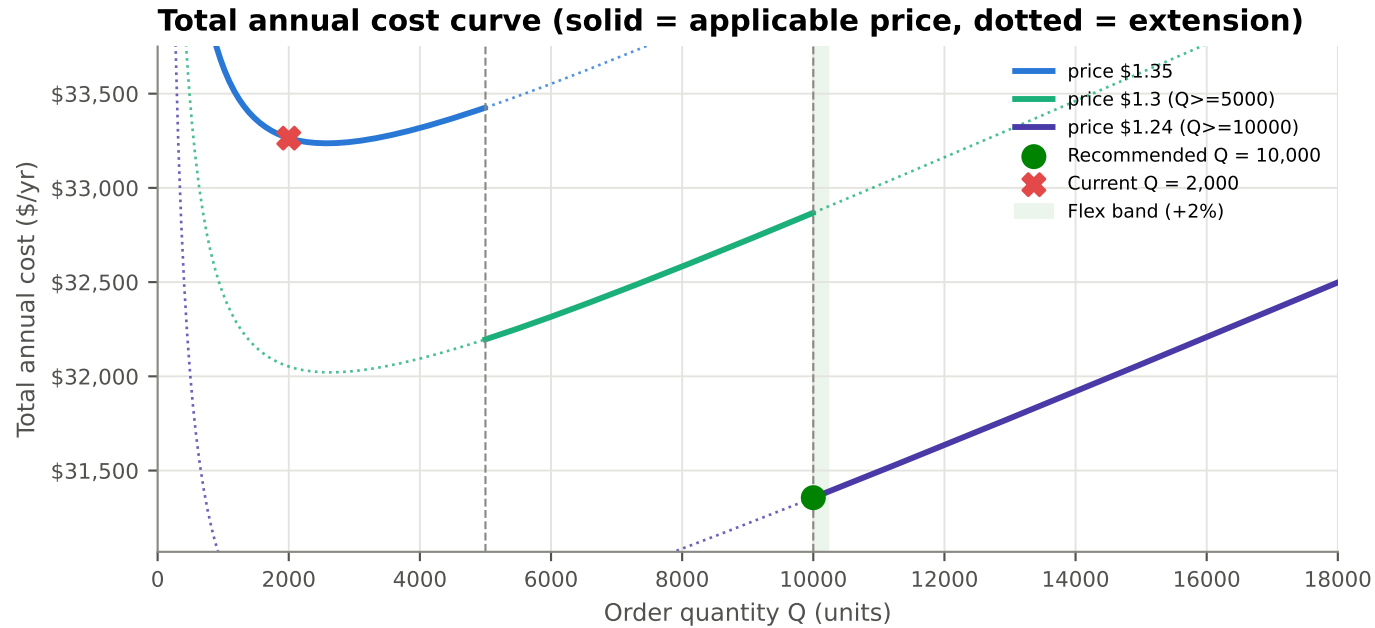
TRC breakdown (per year)



Key figures

Metric	Value
Annual demand D	1,800
Order/setup cost A	\$1,500.00
Base unit cost v0	\$455.000
Carrying rate r	0.22
Production rate m	9,000
Textbook EOQ/EPQ	259.7
Recommended Q	260
Price paid	\$455.000
Orders per year	6.9
Cycle (working days)	36.1
Reorder point	36
TRC (order+carry)	\$20,795
Total annual cost	\$839,795
Case / tier	-
Rules applied	Pack 5
Current Q	450
Annual savings	\$3,223
Current PCP	15.5%

RM-1005 - Ball bearing 6204-2RS [EOQ + discounts]

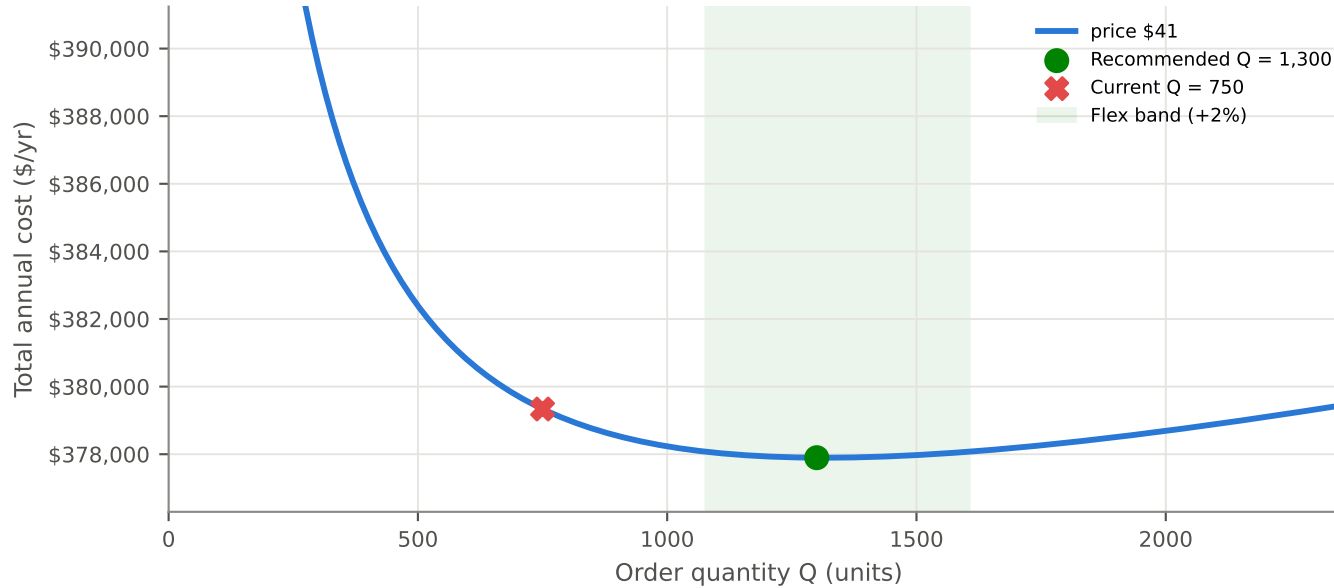


Key figures

Metric	Value
Annual demand D	24,000
Order/setup cost A	\$45.00
Base unit cost v0	\$1.350
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	2,582.0
Recommended Q	10,000
Price paid	\$1.240
Orders per year	2.4
Cycle (working days)	104.2
Reorder point	672
TRC (order+carry)	\$1,596
Total annual cost	\$31,356
Case / tier	Tier 3 of 3
Rules applied	Pack 50
Current Q	2,000
Annual savings	\$1,908
Current PCP	119.5%

FG-3004 - Check valve DN50 [EPQ]

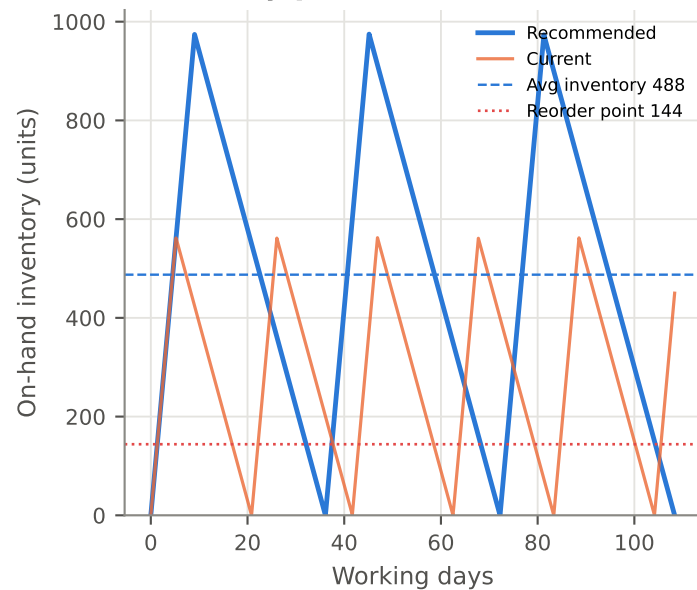
Total annual cost curve (solid = applicable price, dotted = extension)



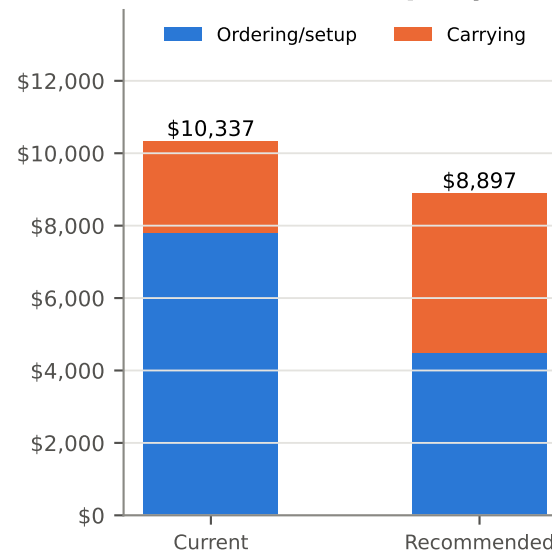
Key figures

Metric	Value
Annual demand D	9,000
Order/setup cost A	\$650.00
Base unit cost v0	\$41.000
Carrying rate r	0.22
Production rate m	36,000
Textbook EOQ/EPQ	1,315.1
Recommended Q	1,300
Price paid	\$41.000
Orders per year	6.9
Cycle (working days)	36.1
Reorder point	144
TRC (order+carry)	\$8,897
Total annual cost	\$377,897
Case / tier	-
Rules applied	Pack 50
Current Q	750
Annual savings	\$1,440
Current PCP	16.2%

Inventory profile over time (sawtooth)

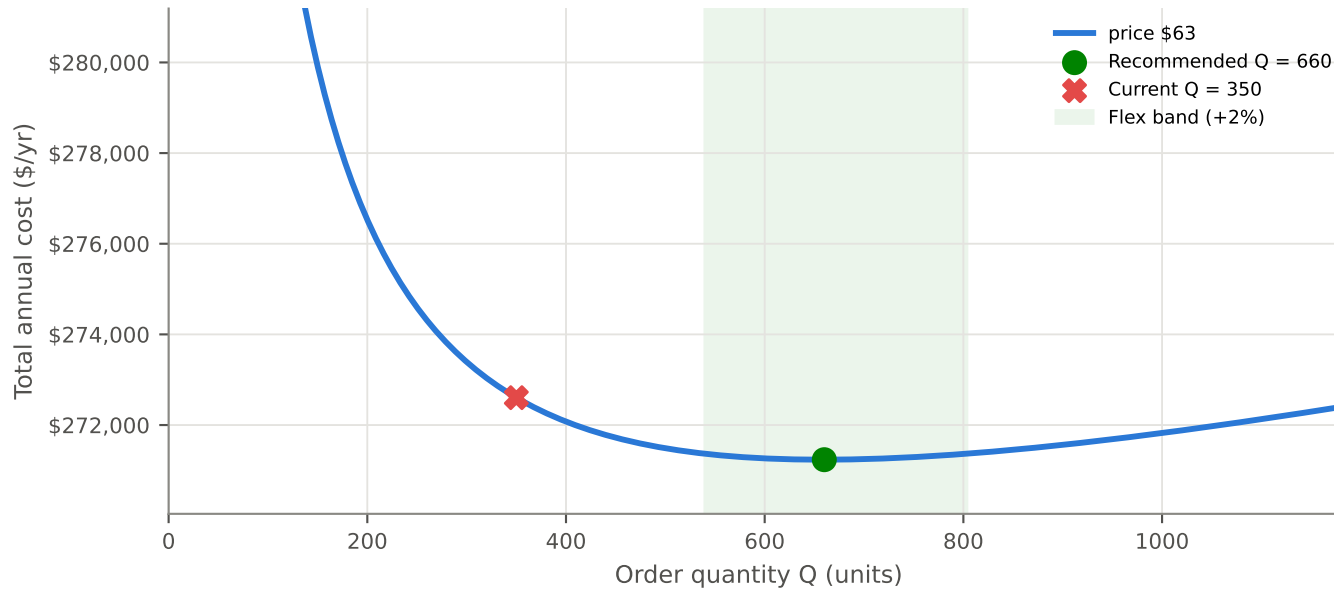


TRC breakdown (per year)

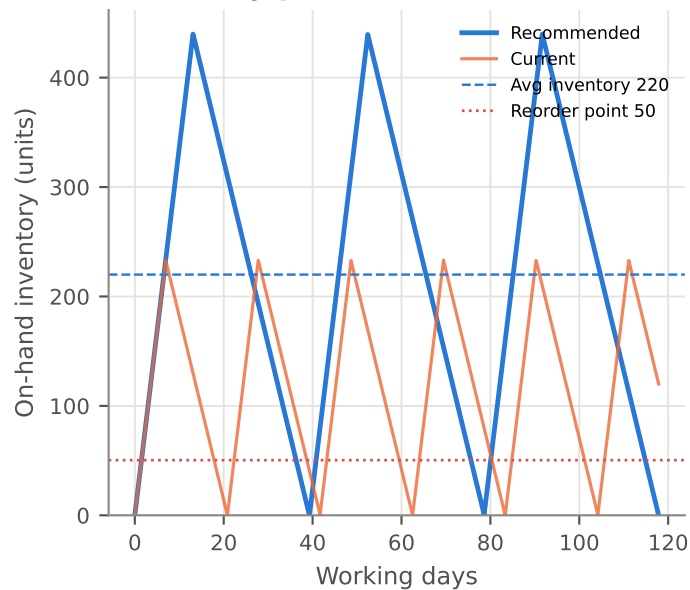


SA-4002 - Welded base frame [EPQ]

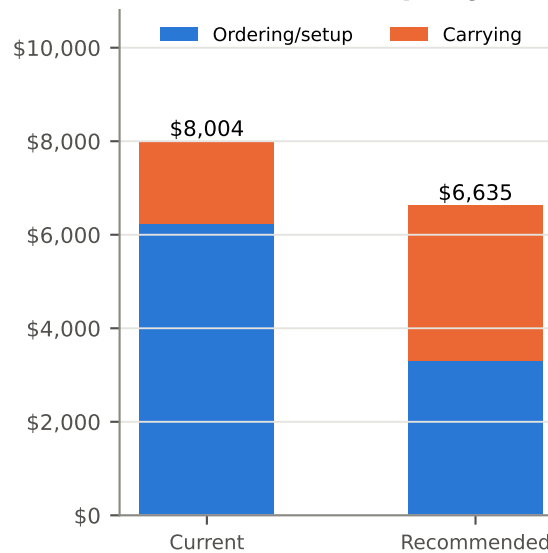
Total annual cost curve (solid = applicable price, dotted = extension)



Inventory profile over time (sawtooth)



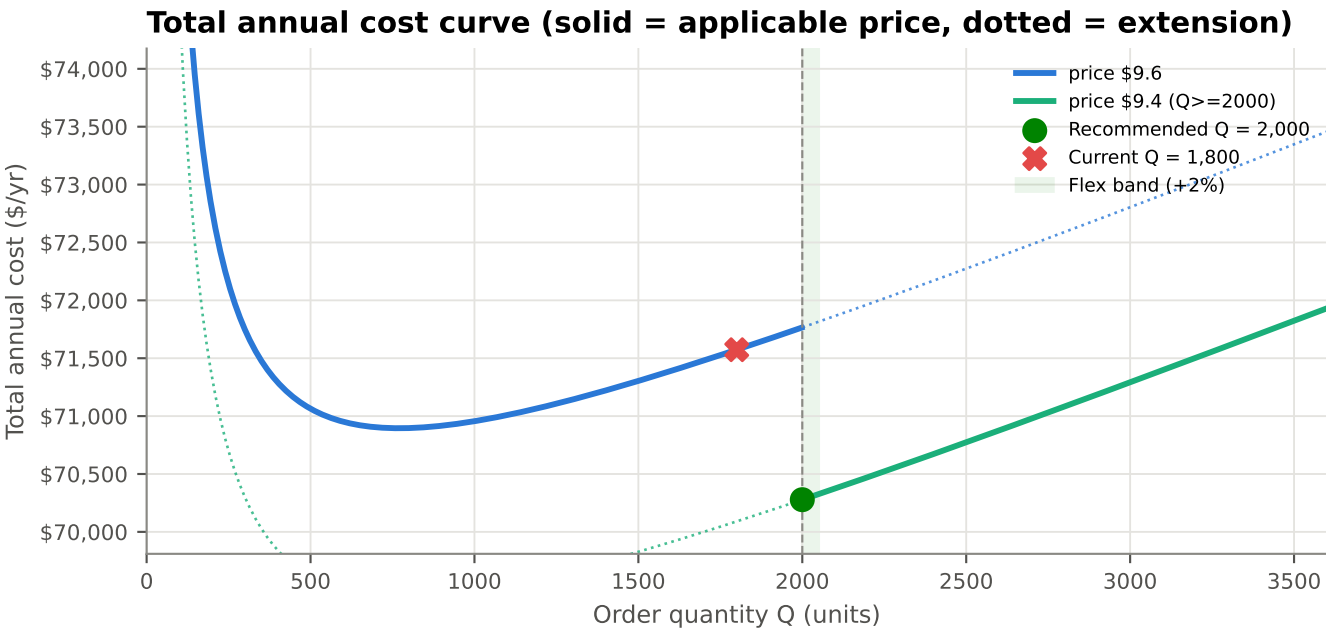
TRC breakdown (per year)



Key figures

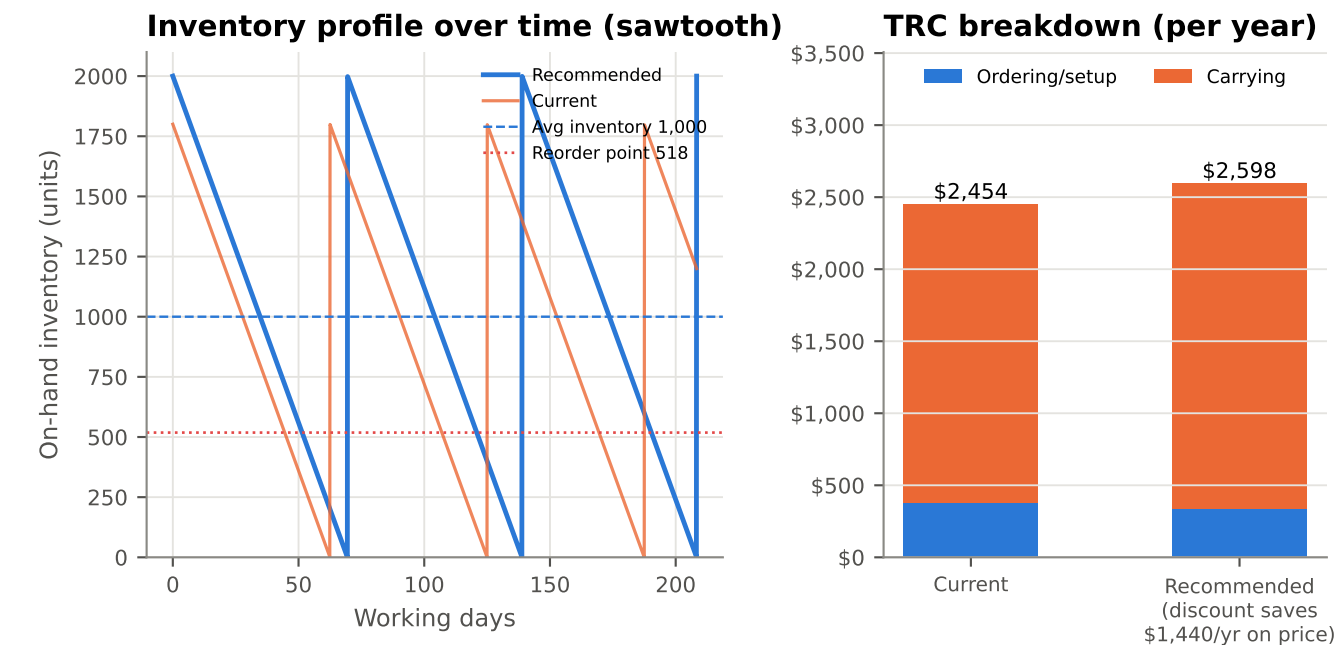
Metric	Value
Annual demand D	4,200
Order/setup cost A	\$520.00
Base unit cost v0	\$63.000
Carrying rate r	0.24
Production rate m	12,600
Textbook EOQ/EPQ	658.3
Recommended Q	660
Price paid	\$63.000
Orders per year	6.4
Cycle (working days)	39.3
Reorder point	50
TRC (order+carry)	\$6,635
Total annual cost	\$271,235
Case / tier	-
Rules applied	Pack 5
Current Q	350
Annual savings	\$1,369
Current PCP	20.6%

RM-1012 - Shaft steel 42CrMo4 (m) [EOQ + discounts]

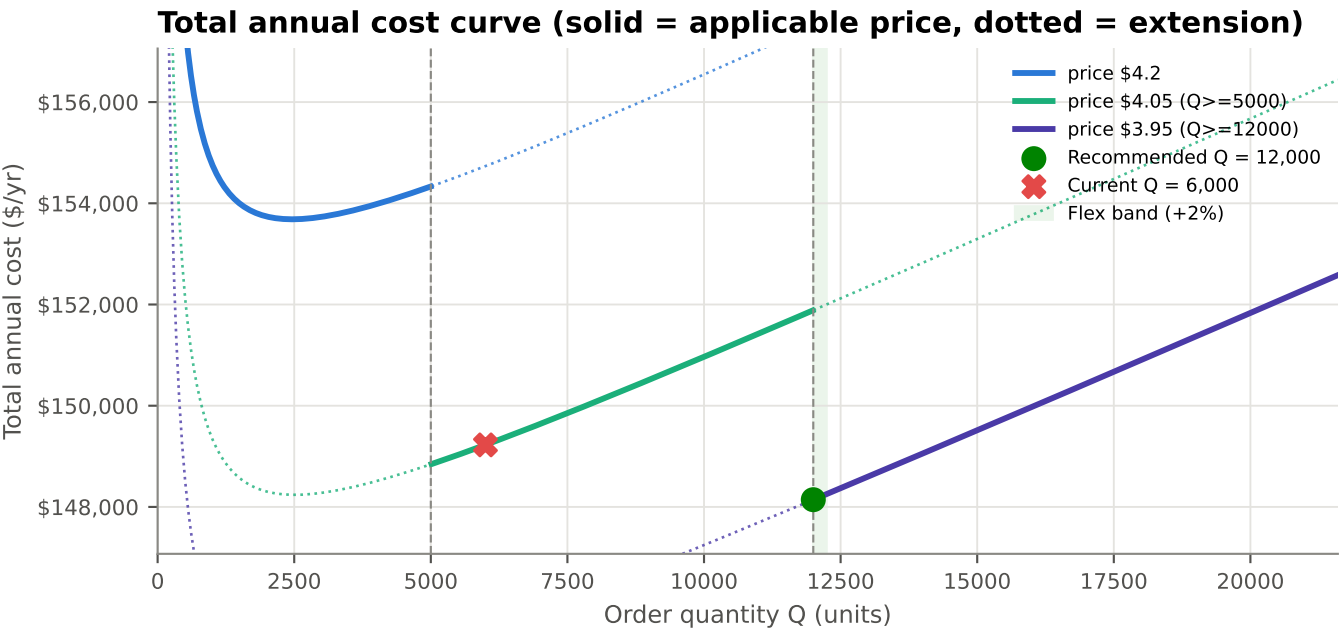


Key figures

Metric	Value
Annual demand D	7,200
Order/setup cost A	\$95.00
Base unit cost v0	\$9.600
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	770.6
Recommended Q	2,000
Price paid	\$9.400
Orders per year	3.6
Cycle (working days)	69.4
Reorder point	518
TRC (order+carry)	\$2,598
Total annual cost	\$70,278
Case / tier	Case 1
Rules applied	Pack 10
Current Q	1,800
Annual savings	\$1,296
Current PCP	49.9%

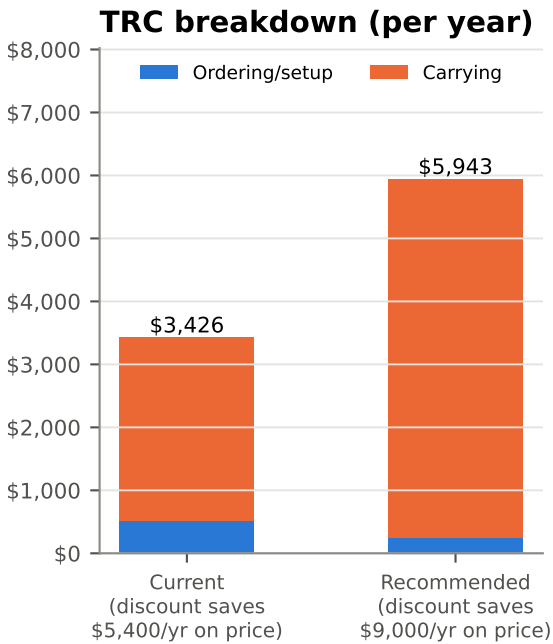
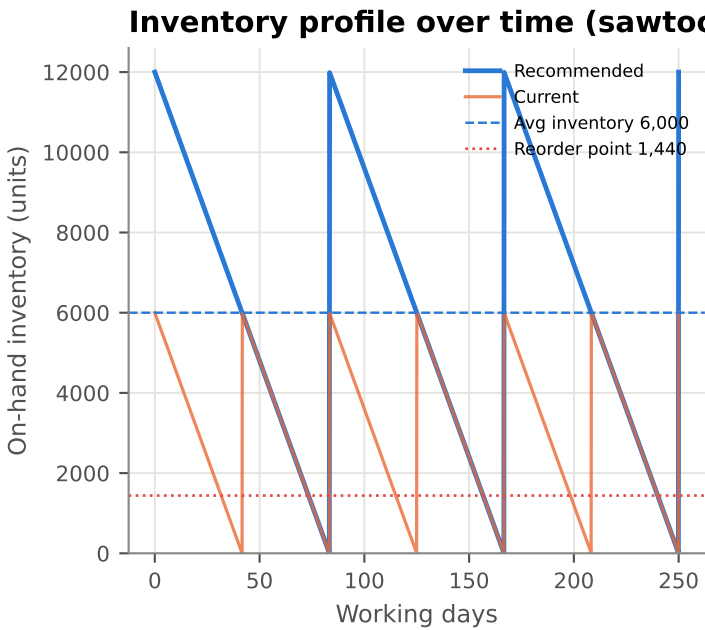


RM-1001 - Steel bar 25mm (m) [EOQ + discounts]

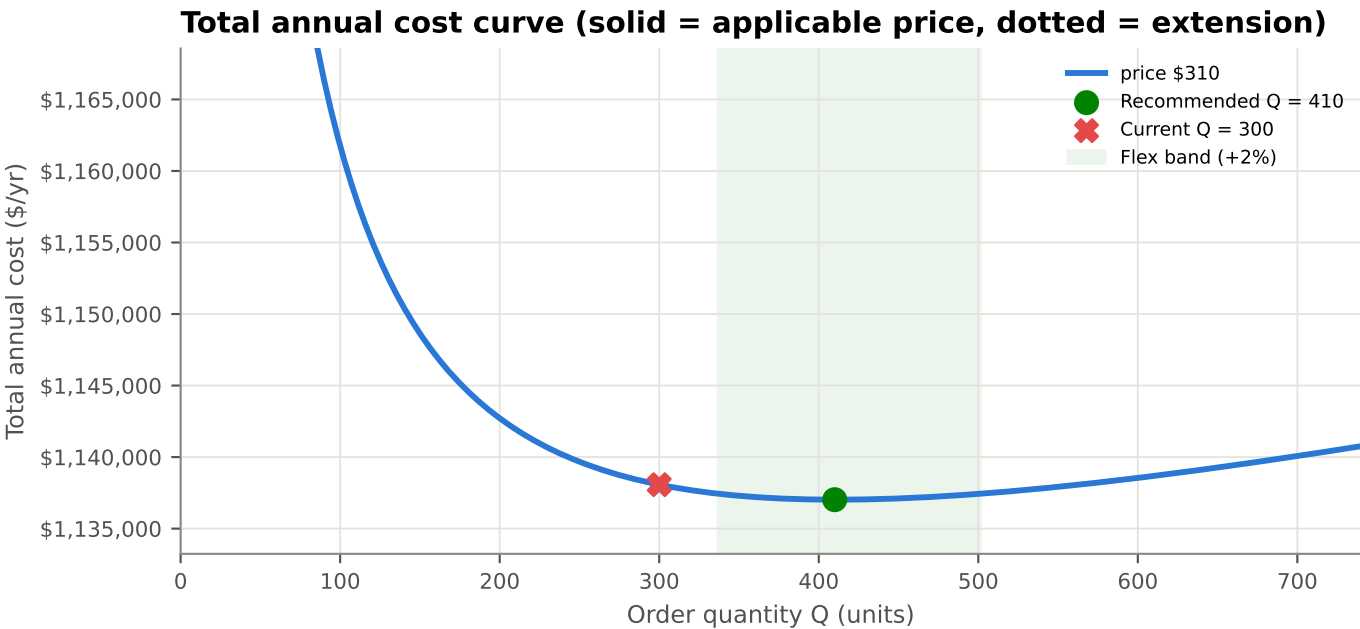


Key figures

Metric	Value
Annual demand D	36,000
Order/setup cost A	\$85.00
Base unit cost v0	\$4.200
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	2,464.0
Recommended Q	12,000
Price paid	\$3.950
Orders per year	3.0
Cycle (working days)	83.3
Reorder point	1,440
TRC (order+carry)	\$5,943
Total annual cost	\$148,143
Case / tier	Tier 3 of 3
Rules applied	Pack 100
Current Q	6,000
Annual savings	\$1,083
Current PCP	18.2%

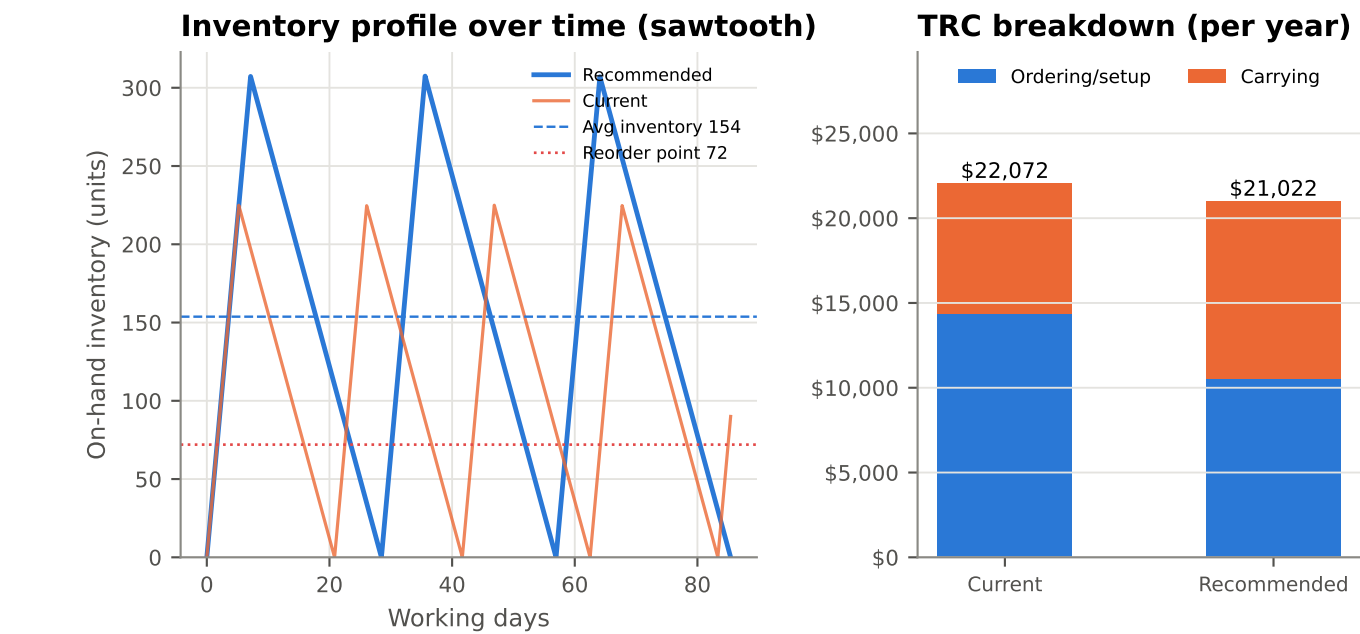


FG-3001 - Centrifugal pump CP-50 [EPQ]



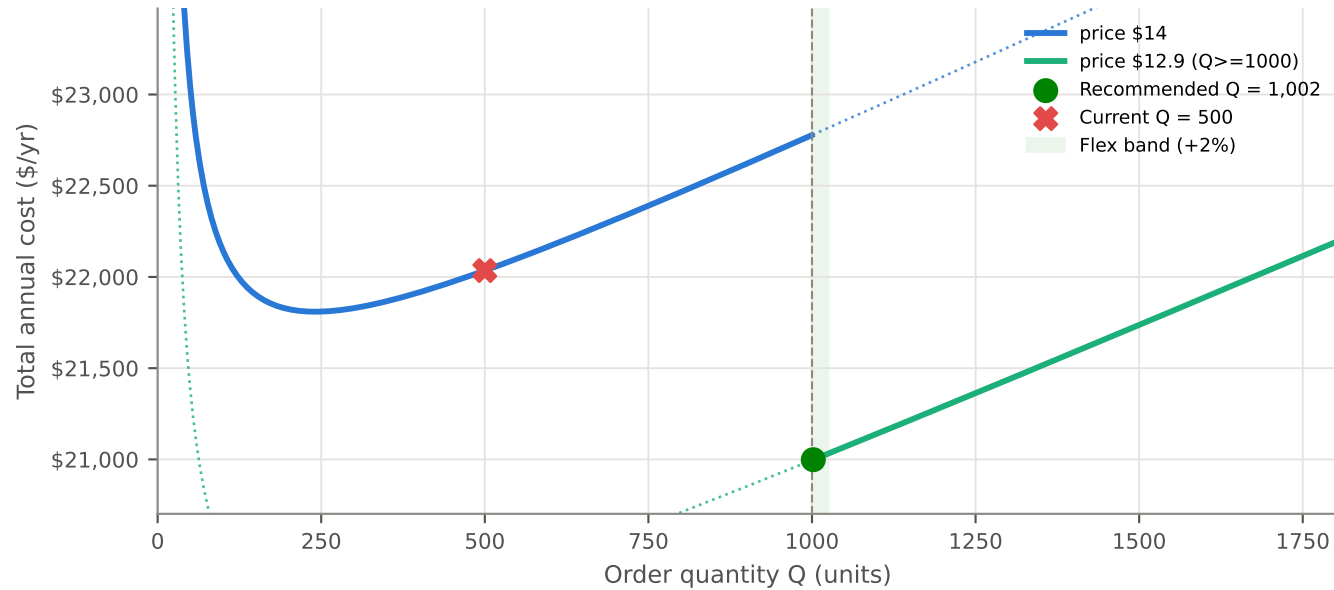
Key figures

Metric	Value
Annual demand D	3,600
Order/setup cost A	\$1,200.00
Base unit cost v0	\$310.000
Carrying rate r	0.22
Production rate m	14,400
Textbook EOQ/EPQ	411.0
Recommended Q	410
Price paid	\$310.000
Orders per year	8.8
Cycle (working days)	28.5
Reorder point	72
TRC (order+carry)	\$21,022
Total annual cost	\$1,137,022
Case / tier	-
Rules applied	Pack 10
Current Q	300
Annual savings	\$1,050
Current PCP	5.0%

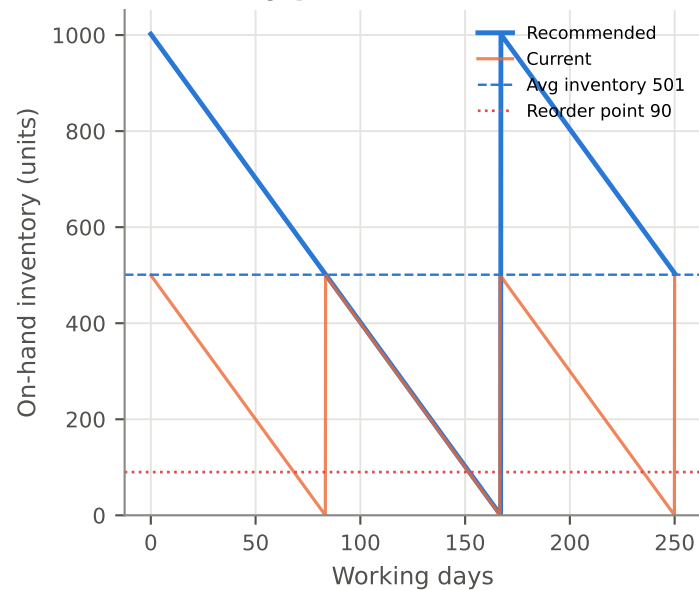


SP-2002 - Spare seal (service) [EOQ + discounts]

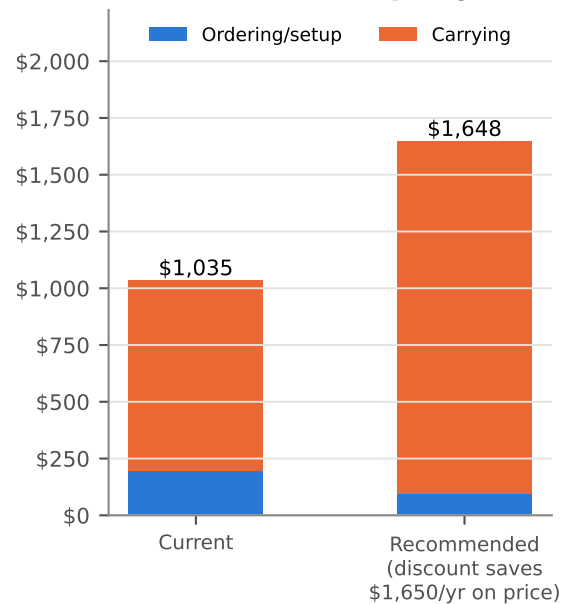
Total annual cost curve (solid = applicable price, dotted = extension)



Inventory profile over time (sawtooth)



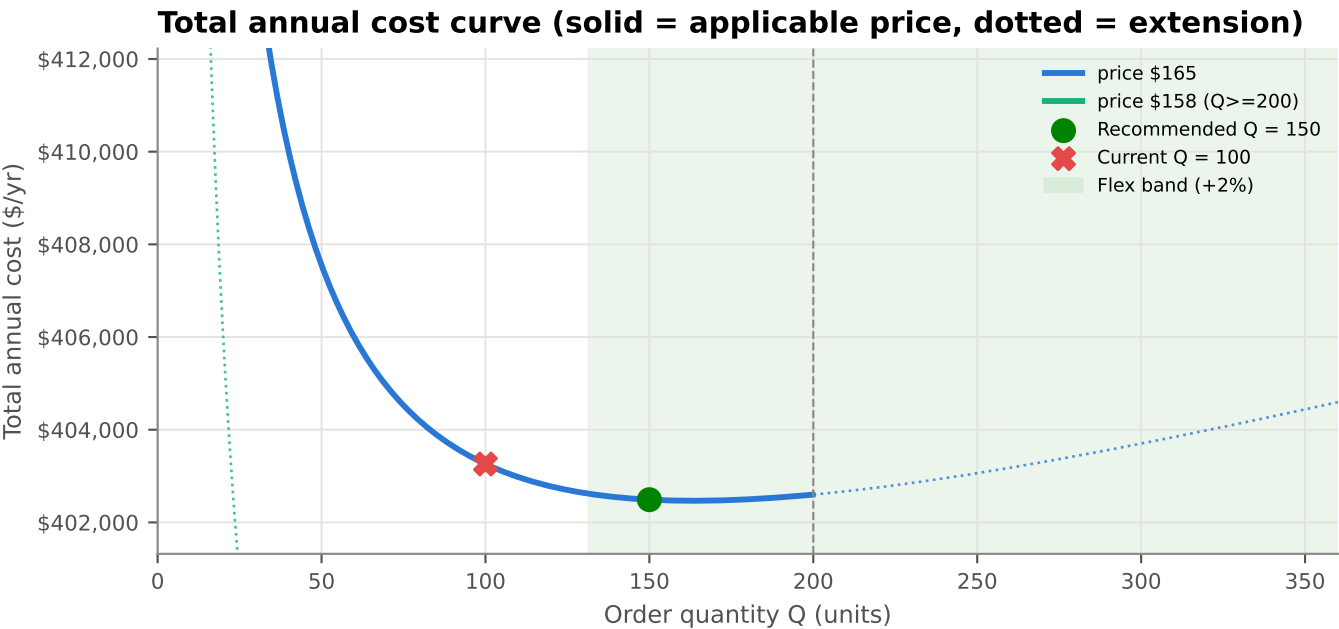
TRC breakdown (per year)



Key figures

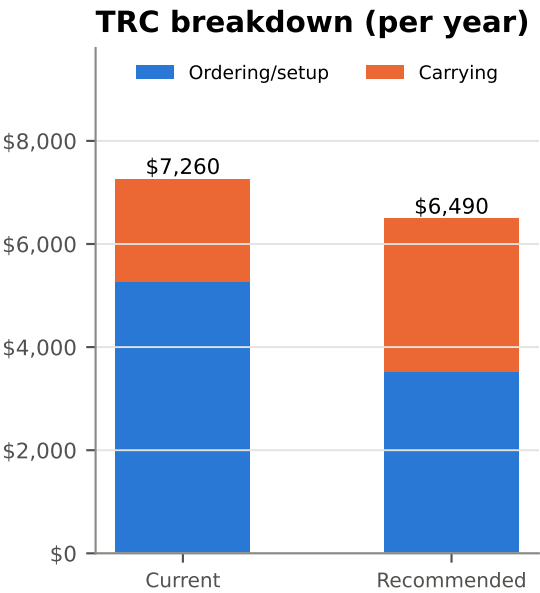
Metric	Value
Annual demand D	1,500
Order/setup cost A	\$65.00
Base unit cost v0	\$14.000
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	240.9
Recommended Q	1,002
Price paid	\$12.900
Orders per year	1.5
Cycle (working days)	167.0
Reorder point	90
TRC (order+carry)	\$1,648
Total annual cost	\$20,998
Case / tier	Case 1
Rules applied	Pack 6
Current Q	500
Annual savings	\$1,037
Current PCP	62.9%

RM-1008 - Electric motor 2.2 kW [EOQ + discounts]



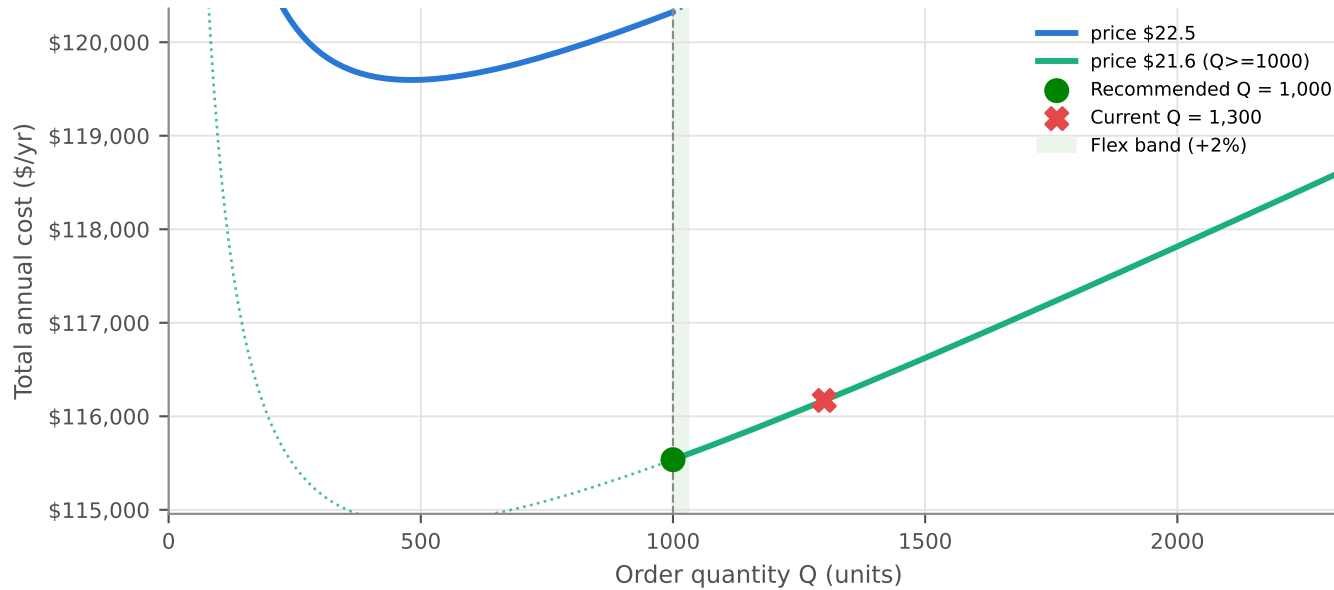
Key figures

Metric	Value
Annual demand D	2,400
Order/setup cost A	\$220.00
Base unit cost v0	\$165.000
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	163.3
Recommended Q	150
Price paid	\$165.000
Orders per year	16.0
Cycle (working days)	15.6
Reorder point	288
TRC (order+carry)	\$6,490
Total annual cost	\$402,490
Case / tier	Case 1
Rules applied	Max qty
Current Q	100
Annual savings	\$770
Current PCP	11.9%

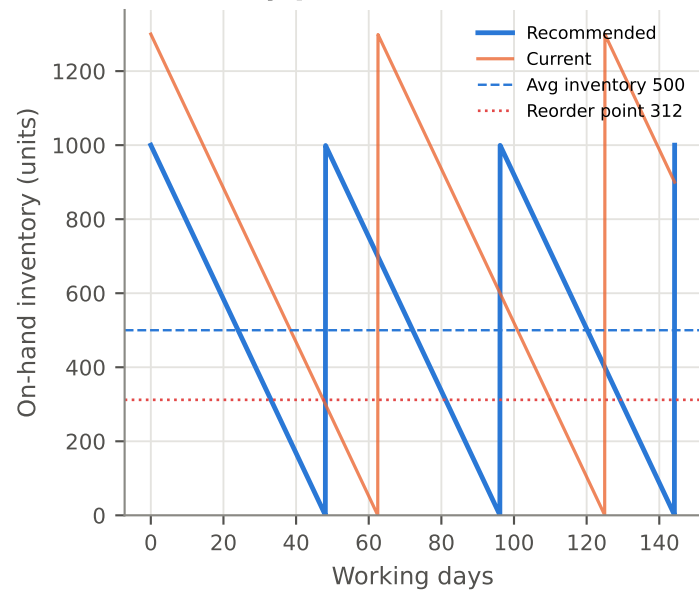


RM-1003 - Bronze impeller blank [EOQ + discounts]

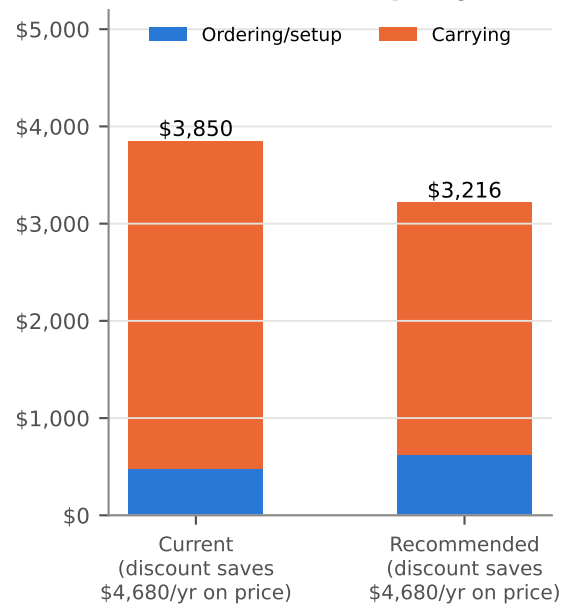
Total annual cost curve (solid = applicable price, dotted = extension)



Inventory profile over time (sawtooth)



TRC breakdown (per year)

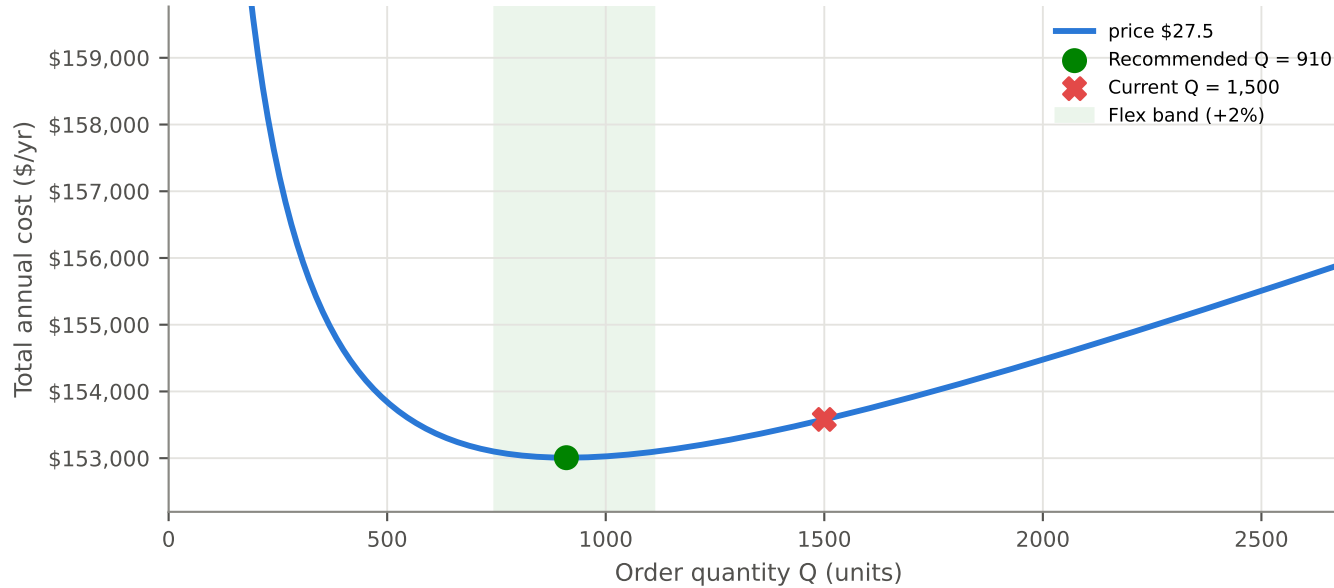


Key figures

Metric	Value
Annual demand D	5,200
Order/setup cost A	\$120.00
Base unit cost v0	\$22.500
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	480.7
Recommended Q	1,000
Price paid	\$21.600
Orders per year	5.2
Cycle (working days)	48.1
Reorder point	312
TRC (order+carry)	\$3,216
Total annual cost	\$115,536
Case / tier	Case 1
Rules applied	Pack 20
Current Q	1,300
Annual savings	\$634
Current PCP	19.7%

SA-4001 - Machined shaft sub-assembly [EPQ]

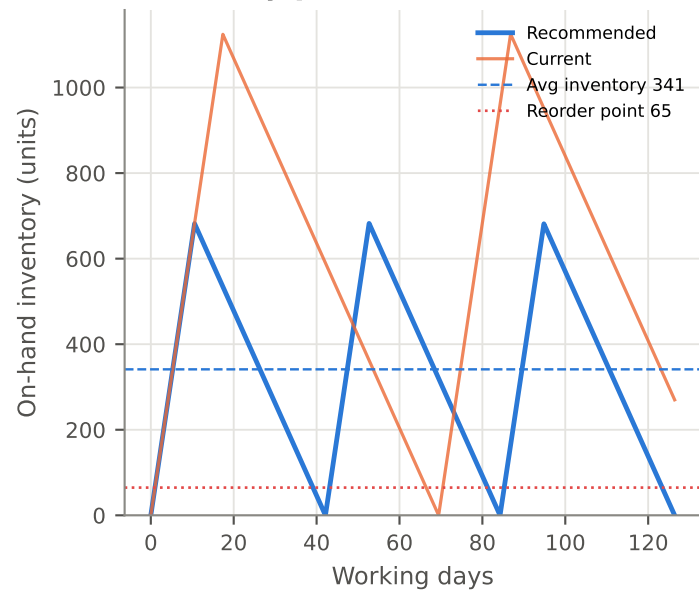
Total annual cost curve (solid = applicable price, dotted = extension)



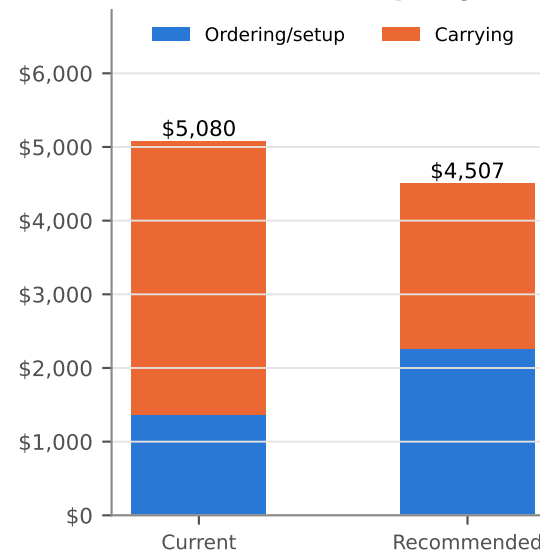
Key figures

Metric	Value
Annual demand D	5,400
Order/setup cost A	\$380.00
Base unit cost v0	\$27.500
Carrying rate r	0.24
Production rate m	21,600
Textbook EOQ/EPQ	910.5
Recommended Q	910
Price paid	\$27.500
Orders per year	5.9
Cycle (working days)	42.1
Reorder point	65
TRC (order+carry)	\$4,507
Total annual cost	\$153,007
Case / tier	-
Rules applied	Pack 10
Current Q	1,500
Annual savings	\$573
Current PCP	12.7%

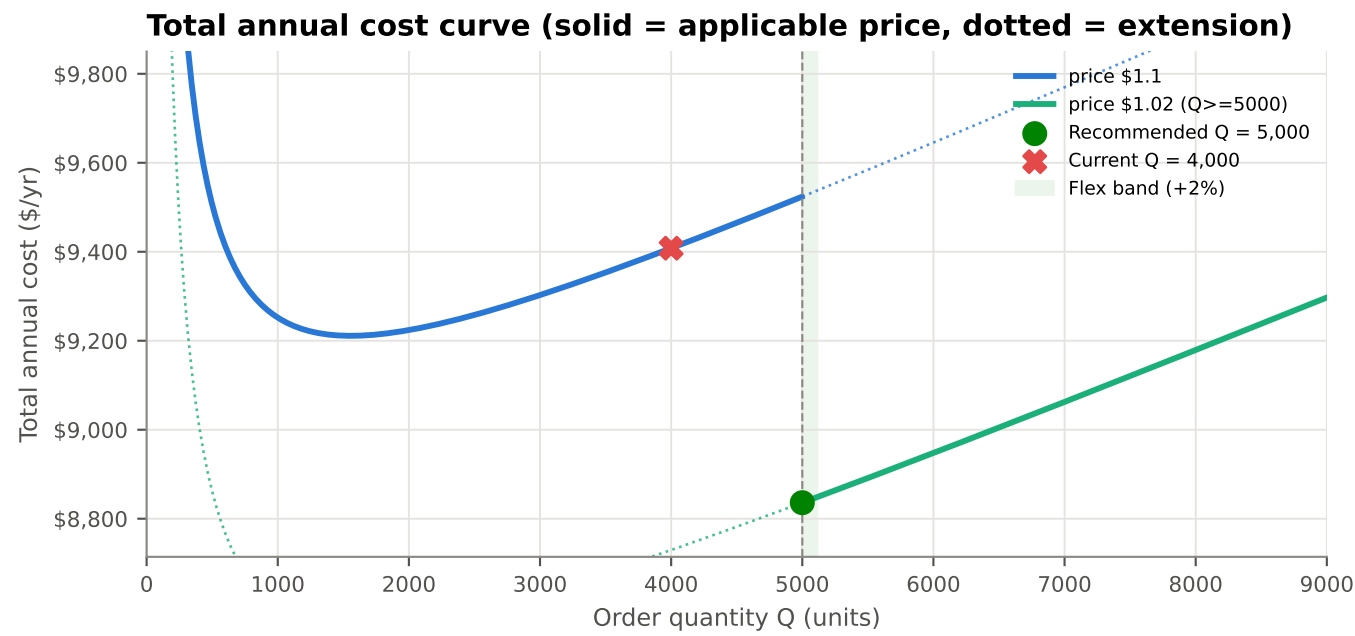
Inventory profile over time (sawtooth)



TRC breakdown (per year)

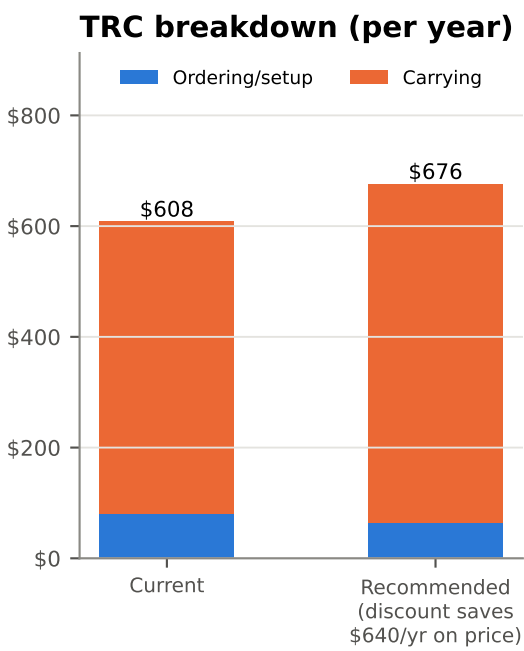


RM-1010 - Packaging carton large [EOQ + discounts]

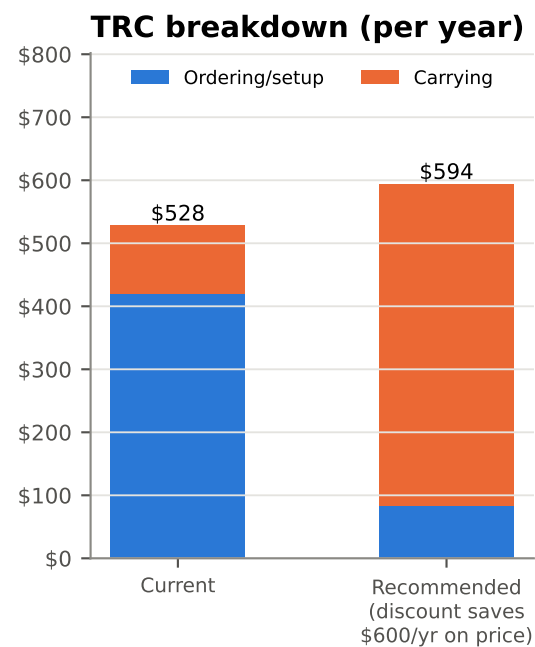
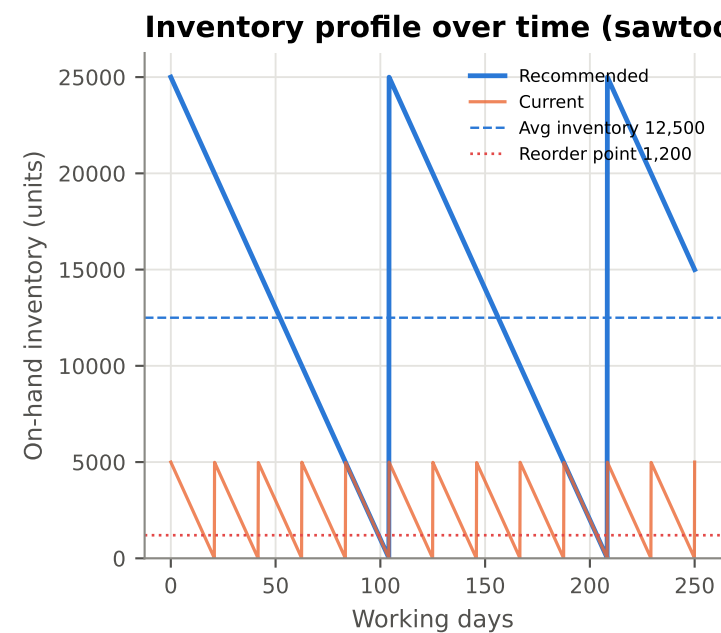
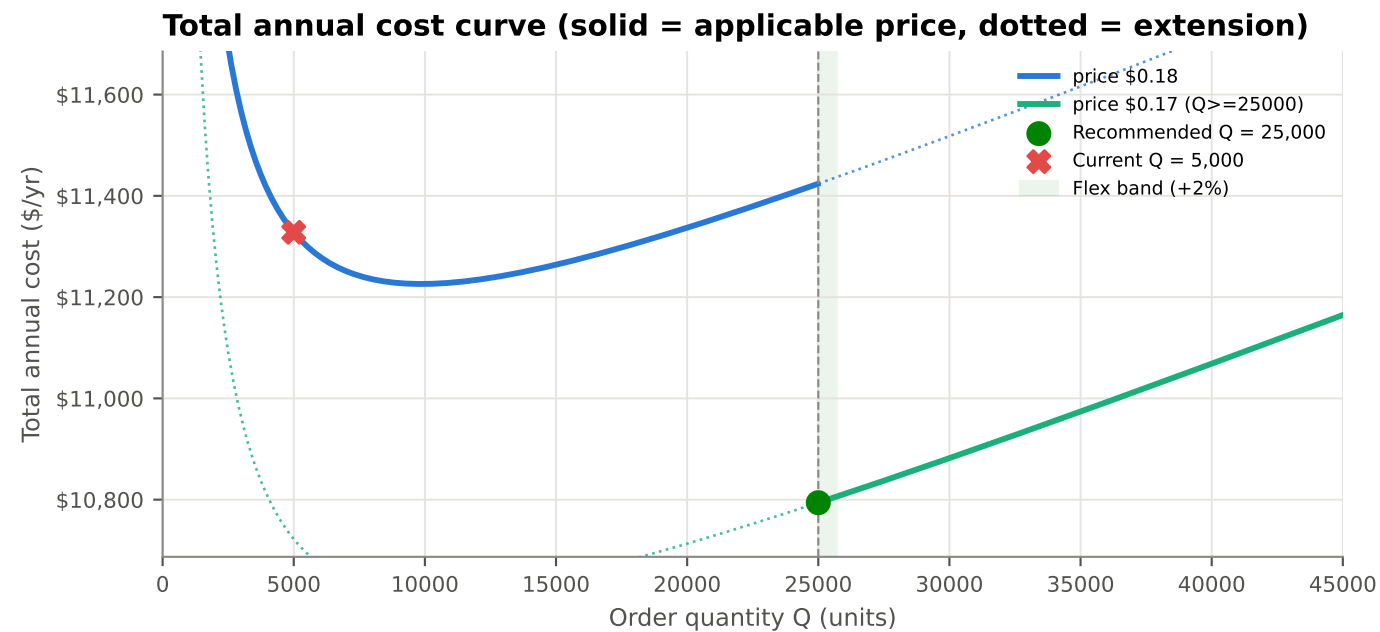


Key figures

Metric	Value
Annual demand D	8,000
Order/setup cost A	\$40.00
Base unit cost v0	\$1.100
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	1,557.0
Recommended Q	5,000
Price paid	\$1.020
Orders per year	1.6
Cycle (working days)	156.2
Reorder point	192
TRC (order+carry)	\$676
Total annual cost	\$8,836
Case / tier	Case 1
Rules applied	Pack 250
Current Q	4,000
Annual savings	\$572
Current PCP	84.6%



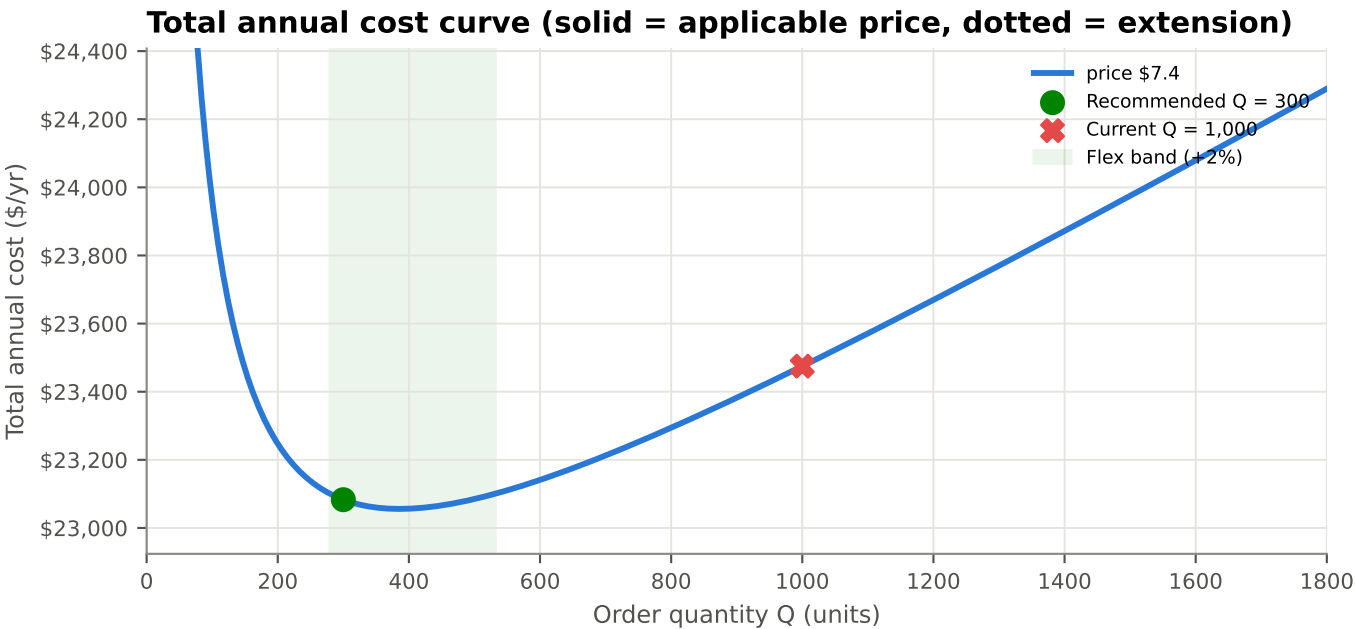
RM-1006 - O-ring set NBR [EOQ + discounts]



Key figures

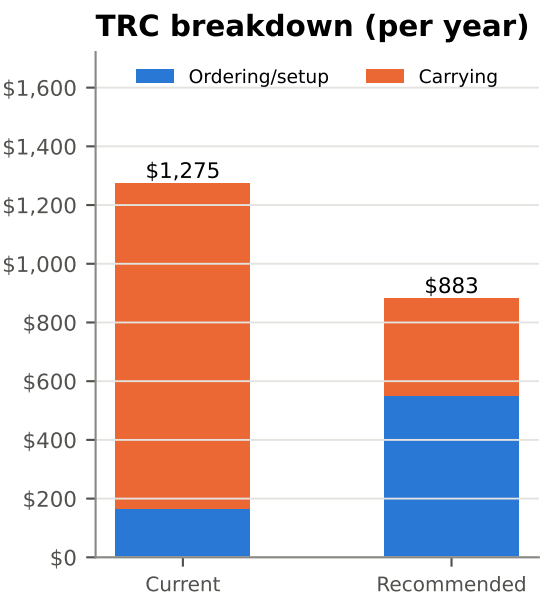
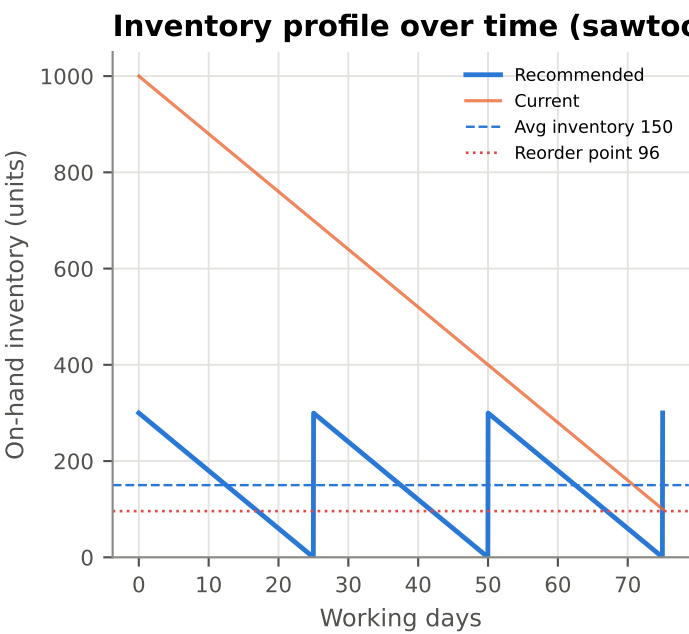
Metric	Value
Annual demand D	60,000
Order/setup cost A	\$35.00
Base unit cost v_0	\$0.180
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	9,860.1
Recommended Q	25,000
Price paid	\$0.170
Orders per year	2.4
Cycle (working days)	104.2
Reorder point	1,200
TRC (order+carry)	\$594
Total annual cost	\$10,794
Case / tier	Case 1
Rules applied	Pack 500
Current Q	5,000
Annual savings	\$534
Current PCP	89.9%

RM-1009 - Paint RAL5010 (litre) [EOQ]



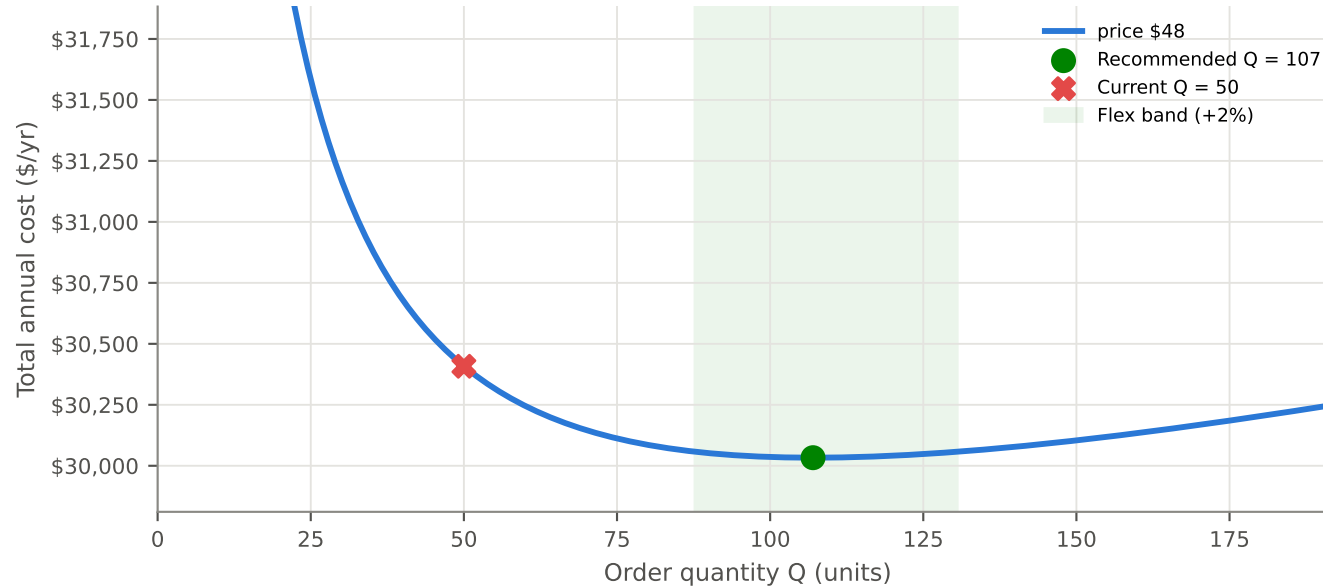
Key figures

Metric	Value
Annual demand D	3,000
Order/setup cost A	\$55.00
Base unit cost v0	\$7.400
Carrying rate r	0.30
Production rate m	-
Textbook EOQ/EPQ	385.5
Recommended Q	300
Price paid	\$7.400
Orders per year	10.0
Cycle (working days)	25.0
Reorder point	96
TRC (order+carry)	\$883
Total annual cost	\$23,083
Case / tier	-
Rules applied	Max qty, Pack 20
Current Q	1,000
Annual savings	\$392
Current PCP	44.4%

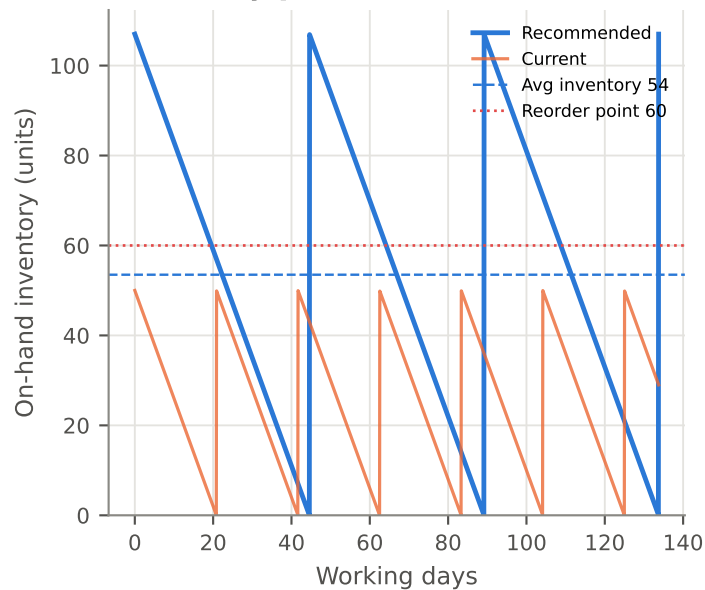


SP-2001 - Spare impeller (service) [EOQ]

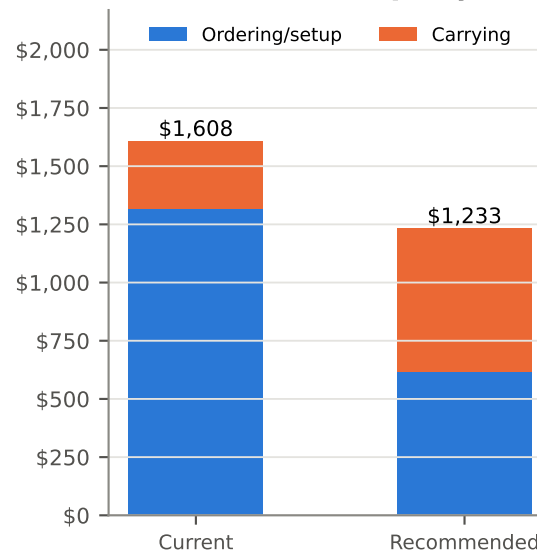
Total annual cost curve (solid = applicable price, dotted = extension)



Inventory profile over time (sawtooth)



TRC breakdown (per year)

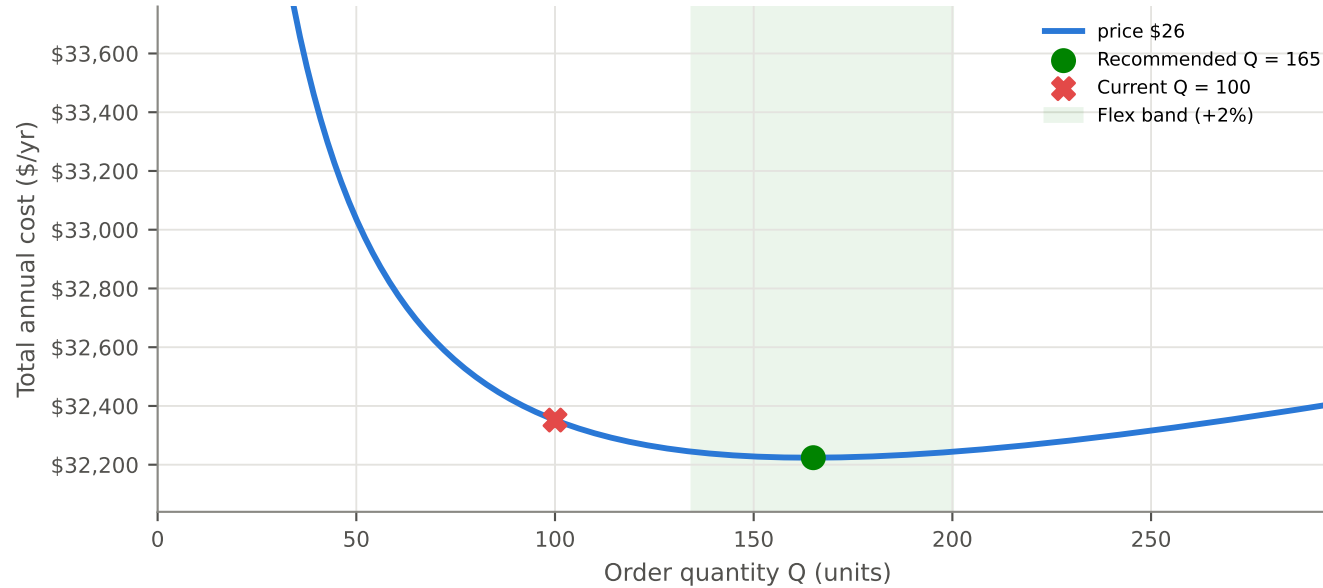


Key figures

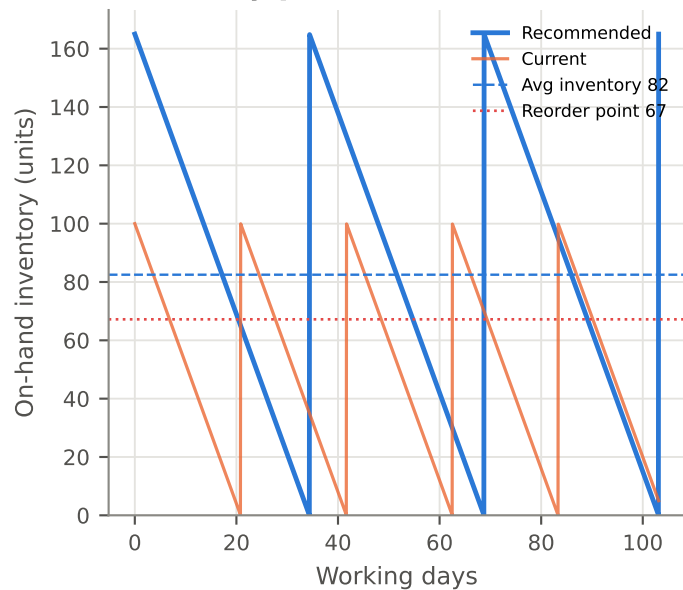
Metric	Value
Annual demand D	600
Order/setup cost A	\$110.00
Base unit cost v0	\$48.000
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	107.0
Recommended Q	107
Price paid	\$48.000
Orders per year	5.6
Cycle (working days)	44.6
Reorder point	60
TRC (order+carry)	\$1,233
Total annual cost	\$30,033
Case / tier	-
Rules applied	none
Current Q	50
Annual savings	\$375
Current PCP	30.4%

RM-1011 - Gasket sheet 1m x 1m [EOQ]

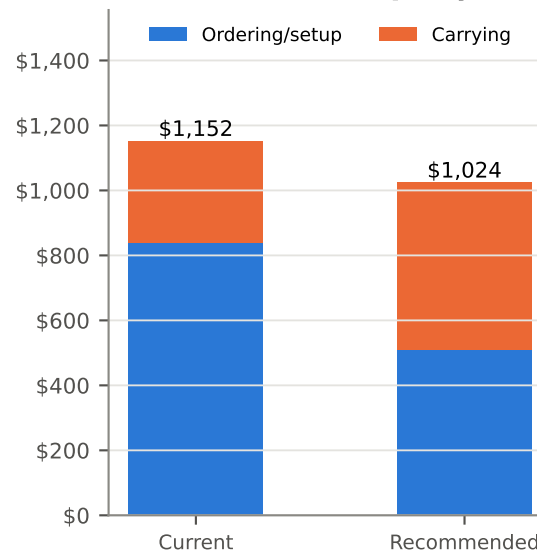
Total annual cost curve (solid = applicable price, dotted = extension)



Inventory profile over time (sawtooth)



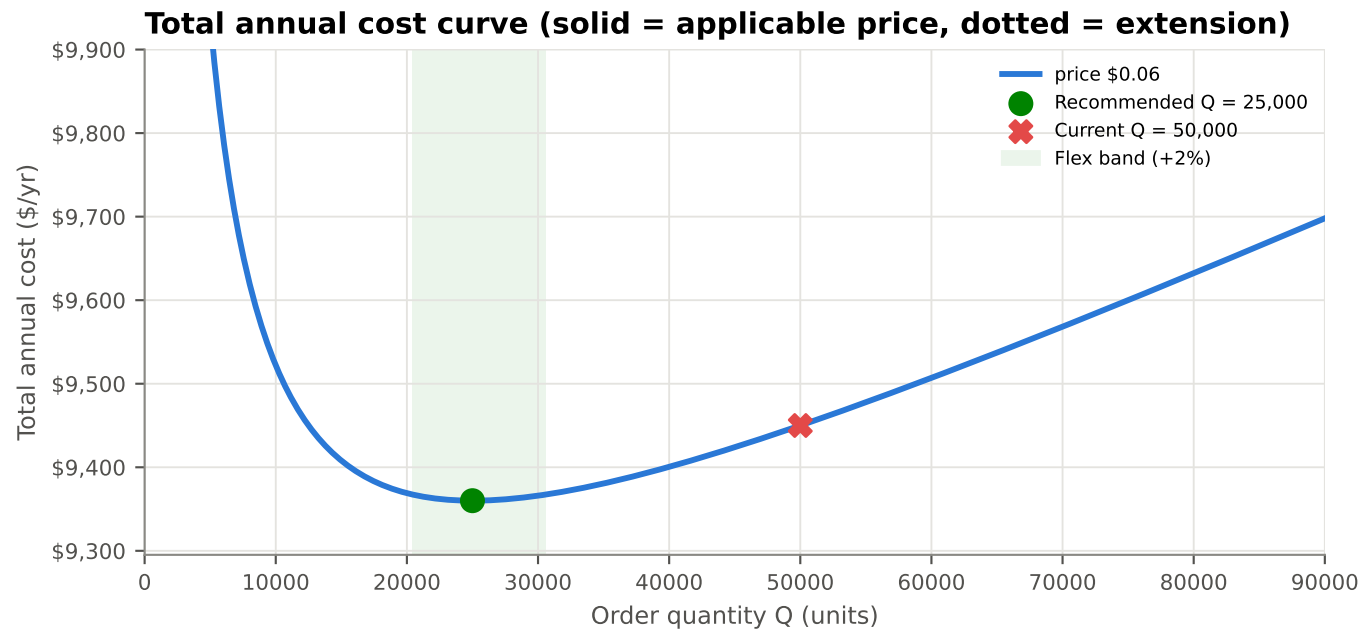
TRC breakdown (per year)



Key figures

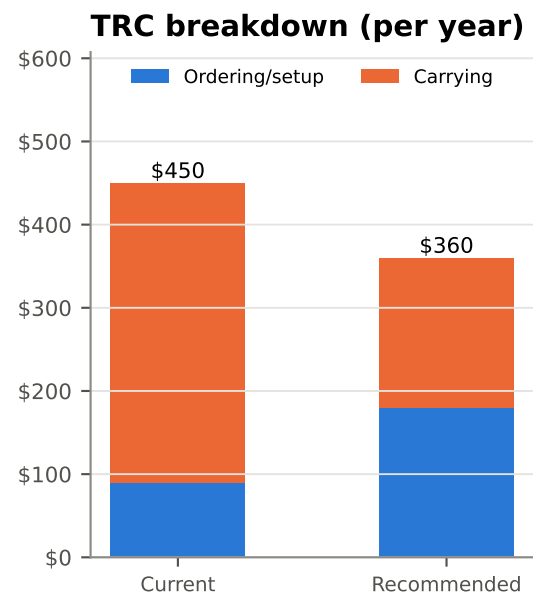
Metric	Value
Annual demand D	1,200
Order/setup cost A	\$70.00
Base unit cost v0	\$26.000
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	164.1
Recommended Q	165
Price paid	\$26.000
Orders per year	7.3
Cycle (working days)	34.4
Reorder point	67
TRC (order+carry)	\$1,024
Total annual cost	\$32,224
Case / tier	-
Rules applied	Pack 5
Current Q	100
Annual savings	\$128
Current PCP	12.5%

RM-1007 - Stainless bolt M10 (box of 1) [EOQ]

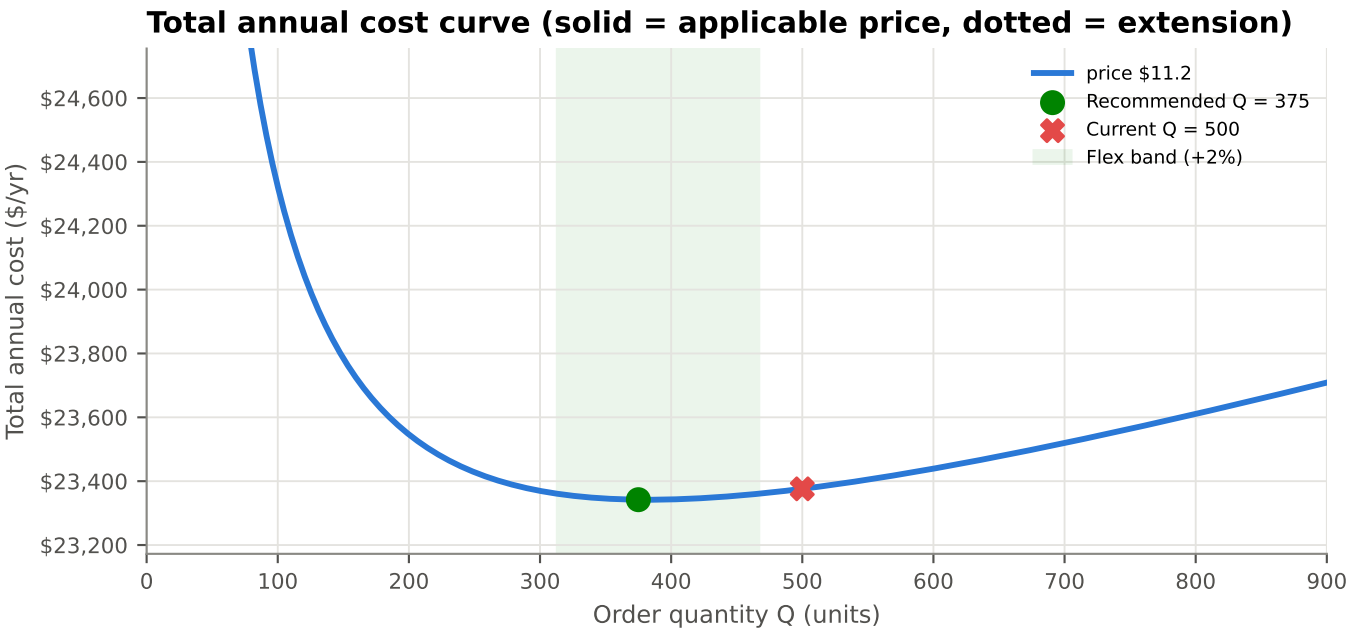


Key figures

Metric	Value
Annual demand D	150,000
Order/setup cost A	\$30.00
Base unit cost v0	\$0.060
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	25,000.0
Recommended Q	25,000
Price paid	\$0.060
Orders per year	6.0
Cycle (working days)	41.7
Reorder point	3,000
TRC (order+carry)	\$360
Total annual cost	\$9,360
Case / tier	-
Rules applied	Pack 1000
Current Q	50,000
Annual savings	\$90
Current PCP	25.0%

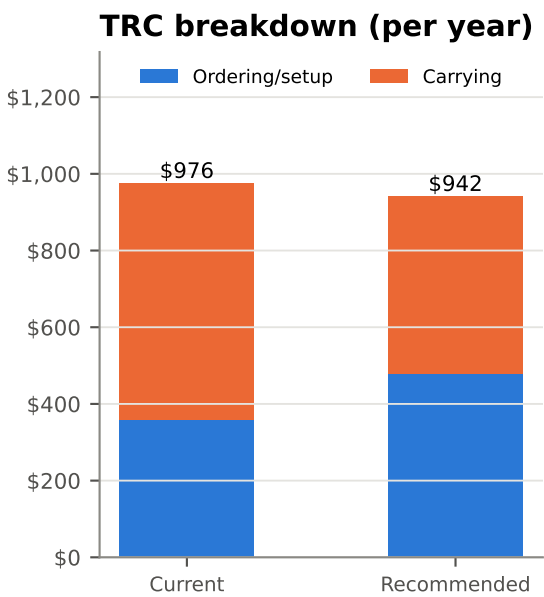


RM-1013 - Copper winding wire (kg) [EOQ]

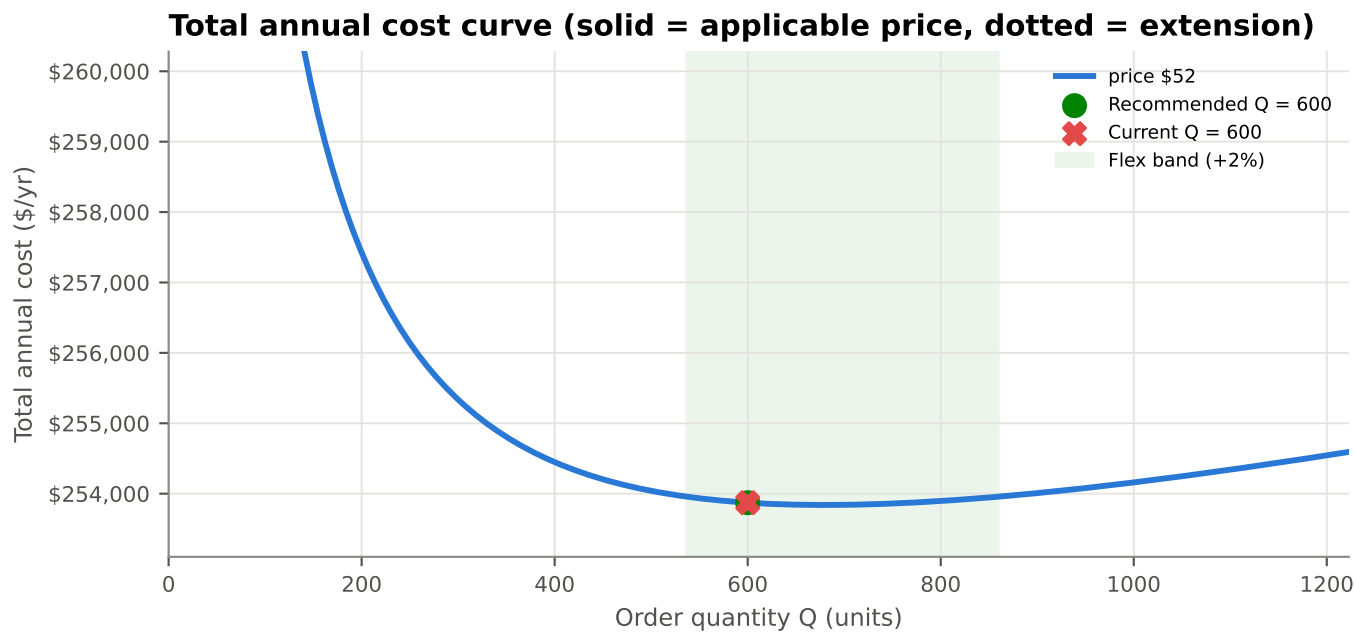


Key figures

Metric	Value
Annual demand D	2,000
Order/setup cost A	\$90.00
Base unit cost v0	\$11.200
Carrying rate r	0.22
Production rate m	-
Textbook EOQ/EPQ	382.2
Recommended Q	375
Price paid	\$11.200
Orders per year	5.3
Cycle (working days)	46.9
Reorder point	160
TRC (order+carry)	\$942
Total annual cost	\$23,342
Case / tier	-
Rules applied	Pack 25
Current Q	500
Annual savings	\$34
Current PCP	3.6%



SA-4003 - Painted housing [EPQ]



Key figures

Metric	Value
Annual demand D	4,800
Order/setup cost A	\$300.00
Base unit cost v0	\$52.000
Carrying rate r	0.24
Production rate m	9,600
Textbook EOQ/EPQ	679.4
Recommended Q	600
Price paid	\$52.000
Orders per year	8.0
Cycle (working days)	31.2
Reorder point	38
TRC (order+carry)	\$4,272
Total annual cost	\$253,872
Case / tier	-
Rules applied	Max qty, Pack 10
Current Q	600
Annual savings	\$0
Current PCP	0.0%

